Actuation-Resolution Language Collapse and Exact Causal Power in a Relational Heat Network

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Abstract

We study the effect of held-input temporal resolution on the complete sampled observation language of a positive, leaky thermal network. The observation alphabet is a quantitative, region-choice-independent projection of source-native EF–FG terminal records, not the native three-color alphabet. For a protected six-node plant with five heater ports and one unheated separator, one input hold per sample makes every word of length two or more impossible at every finite power. Two half-period holds realize every nine-letter word of every length using a causal controller. The exact minimum all-word power is a determinant-defined value $Q_0 \simeq 201.1313457770$, and the online and offline minima coincide. We establish quantitative perturbation bounds and exact power-preserving lifts to arbitrarily large, symmetrically coupled modular thermal networks; controlled heterogeneity and sufficiently weak cross-type links also admit all-word guarantees. Stronger cross-type links lead to new realization obstructions.

Our principal new result closes the all-horizon language phase in a *nontrivially cross-coupled* 12-node network, where each module requests its own symbol. A rigorously certified local critical conductance g_* is the unique root of a three-variable determinant equation in $[0.0054757560, 0.0054757561]$. Throughout this interval, one full-period hold realizes no vector word of length at least two, even without a power bound; its only possible first symbol is $(22, 22)$ for $g \le g_*$, and none is possible for $g > g_*$. With two half-period holds, an explicit *81-action memoryless controller* realizes all 81^n independent vector words for every n, under total phase power 421. Exact rational certificates prove a uniform sampled-guard margin above 0.0293355, and positivity extends the guarantee to specified time-varying cross-conductance perturbations. The proof combines Farkas duals, rigorous matrix-exponential enclosures, a finite min-plus history certificate, and a geometric infinite-tail bound. All statements concern local sampled quantitative records; they do not assert realizability of the joined graph's complete native EF–FG output, intersample safety, or a general-purpose CCR-k computational speedup.

Keywords: positive systems; heat networks; sampled observations; formal languages; Farkas certificates; exact power thresholds; causal control; robustness.

1 Introduction

A continuous physical network may admit individual relational observations without admitting arbitrary sequences of observations under shared actuation constraints. The difference between changing the power budget and changing the temporal resolution of an actuator is particularly important: an additional within-period switch changes the reachable input-to-state map even when no new heater is added.

We provide a fully specified test case in which a one-piece-per-period schedule loses every word of length two, whereas a two-piece schedule realizes the entire free monoid of finite words. The second assertion holds at an *exact*, analytically characterized common-power threshold and without knowing future target symbols.

The basic tools are established: positive-system reachability and linear-programming certificates [9, 10]; positive-state reachability with sign-unrestricted inputs [11]; finite-switch positive reachability on time scales [22]; controllers meeting regular-language specifications [15, 12]; dynamically reconfigurable finite-state control under regular-language specifications [14, 13]; finite-memory temporal control [25]; reachability-based STL synthesis [26]; and abstraction robustness margins [24]. A recent AP-observation automata treatment addresses continuous-time logic and control with interval-wise atomic-proposition information [29]; our guarantees, by contrast, concern only sample times. We therefore make no claim to have introduced these general methodologies. Our contribution is the exact *joint special-model phenomenon* of language collapse, causal power equality, and explicit safe perturbations.

Switching-count and discretization predecessors. The dependence of reachability on switching and sampling deserves explicit priority attribution. Valcher and Santesso [21] study reachability and bounds on switchings in *single-input positive switched systems*; their plant switches among subsystem dynamics, whereas the present comparison fixes one thermal drift matrix $-K$ and refines the number of *held heater inputs within one observation interval*. Bartosiewicz [23] studies when positivity, stability, reachability, and observability survive *time-scale discretization*, including Euler-type discretization; here the sampled maps are the *exact* matrix-exponential images of a continuous-time fixed plant under one or two held inputs. Neither switching-count sensitivity nor sampling sensitivity is therefore claimed as a new general phenomenon. The separate, checkable statement here is a complete language obstruction, a matched exact all-word power optimum and a causal witness for this particular source-defined observation chamber.

The relational predicates follow a protected chamber of the foundational EF–FG ownership and spatial-frame formalism [1, 2] as specialized in the distinct physical joint-realization study [3]. Lemma 2.1 verifies intrinsic witnesses and observer ownership, Proposition 2.2 proves region-independent readout, and Proposition 2.3 classifies every admissible intrinsic region and its possible native color. The three numeric bands are *not* native EF–FG colors (Propositions 2.3 and 2.5), and the projected labels do not require choosing a canonical intrinsic region. The separate CCR computational study [4] concerns a different arithmetic/archival problem. Our controller is executable as an ordinary finite-state lookup; neither a CCR-specific speedup nor a memory optimality claim follows from it.

Comparison with dynamic regular-language controllers. Masciulli and Pola [14, 13] already study finite-state control of continuous-time nonlinear plants when regular-language specifications are only partially known and later revised. Thus the nonanticipative use of the current target letter, finite memory, and regular languages cannot alone be claimed as new. Their central task is synthesis and reconfiguration under changing specifications, with approximation and computational complexity questions; our narrowly specified task is a *sharp input-hold resolution obstruction* on one positive graph, together with the exact physical common-power optimum for realizing every requested word. Our result is a certified special-model phenomenon, not a competing general synthesis method or a faster reconfiguration algorithm. Similarly, the positive-state reachable-space characterization of Guiver, Rebarber and Townley [11] permits sign-indefinite inputs in its central positive-state formulation, whereas every heater input here is constrained to be nonnegative; the reachability questions therefore should not be identified.

Source-native semantics versus a categorical language. The exact source-native objects (seed selection, co-winner basin, observer-record ownership, and intrinsic terminal regions) precede this work in EF–FG I and II [1, 2]. The one-time physical support-function problem also precedes

it [3]. The two terminal colors can vary with the selected intrinsic regions even when the field, full-record query-minus-gate numbers, and all numerical target labels remain unchanged. We do not claim a new EF–FG observer definition, new categorical three-color reachability theorem, or realization of an arbitrary full native tuple. The temporal language and sharp all-word heating threshold concern the declared fixed-query numerical projection only.

Measured-building context and boundary of validation. The broader CCR–k/EF–FG research program has also examined logged building-automation measurements from a 35-zone building and from an operational CSIRO office building with controlled AHU fault interventions [5, 6, 7, 8]. Those separate, exploratory analyses addressed event-sequence reconstruction, the size of finite prediction contexts, and fault-related observability; they did not yield a reliable CCR-specific fault-diagnosis advantage over the strong classical comparators considered there. Crucially, neither archive implements or directly measures the present six- or twelve-node protected sample-guard contract. Thus those field-data investigations support the broader *application motivation*, not the physical experimental validation of the H1/H2 language theorems established below. The results of this article are mathematical theorems supported by exact-arithmetic certificates and finite in-model numerical diagnostics, not field-deployment claims.

Contributions. We specify the physical and observation contract; derive a region-choice-invariant native-record projection and an exact classification of admissible terminal regions; prove one-piece language collapse via 17 nonnegative rational Farkas certificates; determine the sharp two-piece all-word power by an analytic primal–dual basis and a staged invariant controller; establish a parameter-uniform robustness theorem with an explicit power–tolerance region; and prove exact language and online/offline power scaling for arbitrarily many coupled copies under a synchronous, locally evaluated source-native observation contract. For mismatched modules we prove all-size, coupling-strength-independent quantitative H1/H2 persistence and a two-sided power estimate, without claiming an unchanged heterogeneous sharp optimum. The modular lift concerns a structured unbounded family, not arbitrary graphs or global EF–FG semantics on the enlarged graph. Subsequent sections treat independent local words under weak cross-type links, certify a locally unique analytic cross-link threshold, and close the associated H1/H2 vector-word phase for every horizon using an 81-action memoryless controller and 81 two-sample Farkas duals.

2 Plant, semantics, and languages

Let $V = (r, w, p, b, t_1, t_2)$ be ordered as written. Give the undirected edges rw, wp, pb, bt_1, bt_2 conductances $2, 3, 3, \kappa_1, \kappa_2$, respectively, with no other edges. Let $L(\kappa)$ be the weighted graph Laplacian and put

$$K(\kappa) = I_6 + L(\kappa), \qquad \dot{x}(t) = -K(\kappa)x(t) + u(t), \qquad x(0) = 0. \tag{1}$$

The heater u_b is disabled; the five available ports are $P = (r, w, p, t_1, t_2)$. Every held input has nonnegative components and satisfies $\sum_{j\in P} u_j \le Q$. A sample interval of duration τ is subdivided into H equally long constant-input phases. We compare $H = 1$ and $H = 2$. The budget Q bounds each held phase's *power rate*, not its time-integrated energy.

Define the common guard family, evaluated only at sample times $k\tau$, by

$$x_r - x_p \ge 1, \qquad x_p - x_w \ge \frac{5751}{1000}, \qquad x_b - x_w \ge 1, \qquad x_p - x_{t_i} \ge 1 \quad (i = 1, 2). \tag{2}$$

For $i \in \{1, 2\}$ and $s_i \in \{0, 1, 2\}$ impose $x_{t_i} - x_b \in I_{s_i}$, where

$$I_0 = [1/2, 1], \qquad I_1 = [2, 3], \qquad I_2 = [4, 9/2]. \tag{3}$$

The target alphabet is $\Sigma = \{0,1,2\}^2$, ordered lexicographically. For $s = (s_1, s_2)$, let $H_s x \geq b_s$ comprise the five inequalities in (2) and two lower and two upper terminal-band inequalities. We use nine rows per H_s. All constraints are affine *state-difference* guards. The ownership condition $x_p - x_w > 23/4$ follows from the stronger $5751/1000$ inequality. The letter is a *declared quantitative binning* of the region-choice-independent full-record coordinate $y_i - s_i^{\mathrm{gate}} = x_{t_i} - x_b$, not one of the three native polarity/terminal-classification colors. We assert neither these guards at unsampled times nor a characterization of all possible EF–FG source-native tuples (s, U, y, χ).

Lemma 2.1 (Source-native witness and projected observable on the guarded tree)**.** *Fix a sampled field x satisfying* (2)–(3)*, the root observer p, selected-seed threshold $\Delta_{\mathrm{seed}} = 5$, observer interpolation parameter $\theta = 2/5$, and bright-ownership threshold $\Delta_{\mathrm{own}} = 23/4$. Compute primitive extrema, superlevel elders, maximin attachments, and the two observer polarities using the source-native rules of [3]. Then r is the essential bright maximum, p is the sole selected finite bright seed, and the bright and dark observer saliencies at p are $x_p - x_w$ and 0, respectively. Hence p has strict bright ownership. For each terminal t_i, the intrinsic residual-region family admits the connected witness*

$$R_i^{\mathrm{path}} = \{b, t_i\}, \qquad b \in R_i^{\mathrm{path}}, \quad s_i^{\mathrm{gate}} = x_b, \quad L(R_i^{\mathrm{path}}) = x_b, \quad U(R_i^{\mathrm{path}}) = x_{t_i}.$$

At the native query vertex t_i, the complete selected-region record is

$$(s_i^{\mathrm{gate}}, U, y, \chi_{t_i}) = (x_b, x_{t_i}, x_{t_i}, 1), \qquad U - L = x_{t_i} - x_b.$$

Consequently the two numerical band labels are well-defined readings of one source-native joint witness. Proposition 2.2 strengthens this to all *admissible region choices at the same terminal. The labels are not the native dark/neutral/bright classifications.*

Proof. The guards imply $x_r > x_p > x_{t_i} > x_b > x_w$. Hence the only local maxima are r, p, t_1, t_2 and the only local minima are b, w. On the tree the maxima t_i merge into the older component of p at level x_b, while p merges into the older component of r at level x_w. Thus p has finite persistence $x_p - x_w > 5$ and the terminal maxima have persistence $x_{t_i} - x_b \leq 9/2 < 5$; r is essential. For the bright observer at p, the older r attaches through w at level x_w, whereas the seed p attaches at its own birth x_p. Later terminal maxima attach through b below x_p, so no extra bright observer record occurs. Under the negative field, the minima w and b both attach to the observer through a path whose largest original-field value is x_p; the younger dark atom therefore fails the strict record-attachment test. The saliencies are as stated. For the path $p - b - t_i$, the maximin terminal gate is $s_i^{\mathrm{gate}} = \min\{x_p, x_b, x_{t_i}\} = x_b$. The vertex p is forced onto the peak side, t_i onto the terminal side, and b lies in the equal-attachment tie zone, so $\{b, t_i\}$ is a connected intrinsic admissible terminal region. The construction uses the *same* field and elder choice for $i = 1, 2$; the source-native definition imposes no disjointness on the two regions, which intersect at b. Since $x_{t_i} > x_b$, its floor, cap, gate and native query tuple have the displayed values. Finally $x_p - x_w \geq 5751/1000 > 23/4$ yields strict bright ownership. □

Proposition 2.2 (Region-choice invariance of the native numerical projection)**.** *Under the conditions of Lemma 2.1, let R_i be* any *intrinsic residual terminal region for t_i that contains the common separator b. Its full source-native record evaluated at the terminal is*

$$\mathrm{ResRel}_{p,t_i,t_i}(x, R_i) = (x_b, U(R_i), x_{t_i}, 1).$$

Thus the numerical quantity

$$\pi_i\big(\,\mathrm{ResRel}_{p,t_i,t_i}(x, R_i)\big) := y_i - s_i^{\mathrm{gate}} = x_{t_i} - x_b$$

is independent of R_i. Consequently the nine-symbol labeling from (3) *does not require the prior choice of a particular admissible region. In contrast, the cap width $U(R_i) - L(R_i)$ and the native terminal color may depend on the selected region.*

Proof. The gate $s_i^{\mathrm{gate}} = \kappa_{x|\mathcal{B}_p}(p, t_i) = x_b$ depends on the field and terminal but not on a residual region. At the fixed query point t_i, the value $y_i = x_{t_i}$ is also independent of that region, and every admissible region contains the forced terminal, so $\chi_{t_i} = 1$. Subtracting these two native record coordinates gives $\pi_i = x_{t_i} - x_b$ for every allowed R_i. For noninvariance of the cap and color, take the exact admissible field in vertex order

$$x = (38, 30, 37, 31, 63/2, 35)$$

with labels 02. For terminal t_1, both $R_1^{\mathrm{path}} = \{b, t_1\}$ and $R_1^{\max} = \{b, t_1, t_2\}$ are admissible intrinsic regions. The first has $(L, U, y) = (31, 63/2, 63/2)$, cap width $1/2$, and a bright terminal query at zero native margin. The second has $(L, U, y) = (31, 35, 63/2)$, cap width 4, and a dark terminal query since $63/2 < (31 + 35)/2 = 33$. Both records nevertheless yield $y - s = 1/2$. This also demonstrates why the nine numerical labels must not be equated with native color labels. □

Proposition 2.3 (Complete intrinsic-region and native-color classification)**.** *For the protected tree of Lemma 2.1, the rooted co-winner basin of the selected seed p is $\mathcal{B}_p = \{w, p, b, t_1, t_2\}$. For $i \in \{1, 2\}$ and $j \neq i$, the* complete *intrinsic terminal-region family is*

$$\mathcal{R}_{p,t_i}(x) = \big\{\{t_i\},\ \{b, t_i\},\ \{b, t_i, t_j\}\big\}.$$

Exactly two of these regions satisfy the common-separator query $b \in R_i$. Put $g_i = x_{t_i} - x_b > 0$ and take the original terminal classification margin $\Delta_{\mathrm{term}} = 0$. For the smaller region $R_i^- = \{b, t_i\}$ the source-native terminal color at the fixed query vertex t_i is always bright*. For the larger region $R_i^+ = \{b, t_i, t_j\}$ it is* bright*,* neutral*, or* dark *according as $g_j < 2g_i$, $g_j = 2g_i$, or $g_j > 2g_i$, respectively. The region without b, namely $\{t_i\}$, has native color* neutral *but does not satisfy the common-separator query. In all cases the full-record coordinate $y_i - s_i^{\mathrm{gate}} = g_i$ is unchanged.*

Proof. The only selected finite bright seed is p and the essential older seed is r. On the tree, the seed attachments satisfy $\kappa_x(p, v) > \kappa_x(r, v)$ for $v = p, b, t_1, t_2$, equality for $v = w$, and the reverse inequality for $v = r$. Thus the connected co-winner basin containing p is precisely $\mathcal{B}_p = V \setminus \{r\}$. In this basin the gate $s_i = \kappa_{x|\mathcal{B}_p}(p, t_i) = x_b$. The two maximin functions $\alpha(v) = \kappa_{x|\mathcal{B}_p}(p, v)$ and $\gamma_i(v) = \kappa_{x|\mathcal{B}_p}(t_i, v)$ have the following classes: $\alpha(p) > s_i$ makes p forced peak-side, $\gamma_i(t_i) > s_i$ makes t_i forced terminal-side, and $\alpha = \gamma_i = s_i$ holds exactly at the two eligible tie vertices b, t_j. The other basin vertex w has both values strictly below s_i, hence is in neither the forced terminal component nor the tie zone. Every intrinsic admissible terminal region must therefore contain t_i, exclude p, w, and may include b, t_j only subject to connectedness. The three listed subsets are exactly the connected possibilities.

At terminal query $x = t_i$ each admissible region has $y = x_{t_i}$, $\chi = 1$, and the same gate $s_i = x_b$. For a separator-containing region its floor is $L = x_b$, whereas its cap is either $U = x_b + g_i$ or $U = x_b + \max(g_i, g_j)$. At zero terminal margin the original EF–FG midpoint rule compares y with $(L + U)/2$: values above, at, and below the midpoint are bright, neutral and dark, respectively. The smaller region has $y = U > L$ and is always bright. The larger region is bright exactly when $2g_i > \max(g_i, g_j)$, equivalent to $g_j < 2g_i$; equality and reverse inequality give the other two colors. For the singleton $L = U = y$, hence neutral. Finally $y - s_i = g_i$ does not depend on any region choice. □

Corollary 2.4 (Nine-band possibilities for the maximal region)**.** *For $s_i \in \{0, 1, 2\}$ the band interval of g_i is $I_0 = [1/2, 1]$, $I_1 = [2, 3]$, $I_2 = [4, 9/2]$. Writing B, N and D for the three native colors possible under R_i^+, the* complete *possibilities are*

	$s_j = 0$	$s_j = 1$	$s_j = 2$
$s_i = 0$	B, N	N, D	D
$s_i = 1$	B	B	B, N, D
$s_i = 2$	B	B	B

.

At the smaller region R_i^-, the native color remains B in every cell. Thus the nine numerical labels are not equivalent to the intrinsic categorical color records.

Proof. The affine difference $g_j - 2g_i$ ranges over the exact interval $[\inf I_j - 2\sup I_i,\ \sup I_j - 2\inf I_i]$. Applying Proposition 2.3 to its negative, zero and positive values gives precisely the table; the interval endpoints are attainable. □

Proposition 2.5 (The three numerical levels do not encode three native colors)**.** *Assume zero terminal-classification margin in the native EF–FG threshold rules. For each of the nine projected letters, the source-native color of the query at $x = t_i$* when the minimal region R_i^{path} is chosen *is* bright *for both terminals, whereas the color of the query at $x = b$ is* dark*. The complete native query record evaluated at $x = t_i$, however, retains the numerical band information through $y_i - s_i^{\mathrm{gate}} = x_{t_i} - x_b$.*

Proof. The native zero-margin thresholds are $T^+ = T^- = (L + U)/2$. Since $U = x_{t_i} > x_b = L$, at the terminal $y_i = U > T^+$, so its native color is bright regardless of the value of $U - L$. At b the query value is $y = x_b = L < T^-$, so it is dark. In the full native record at t_i both y_i and s_i^{gate} are retained, and their difference determines the declared numerical band. □

Remark 2.6 (Distinction between the native output and the tested language)**.** *The existential predicates $E_i(F; b)$ assert existence of intrinsic regions containing b. The nine-letter observable adds only the three closed, disjoint numerical intervals for the region-independent native-record projection $y_i - s_i^{\mathrm{gate}}$ at query vertex t_i. Unlike $U - L$, this projection needs no prescribed region-selection rule. Its quantization is an* extra output contract*, not a native three-class color. The existence or exact control power of this projected language is not a theorem about every full source-native output tuple.*

For fixed H, Q, let $\mathcal{L}_n(H, Q)$ consist of words $s_1 \cdots s_n$ for which some admissible input schedule makes $H_{s_k} x(k\tau) \geq b_{s_k}$ for each $1 \leq k \leq n$. Write $\mathcal{L}_n(H, \infty) = \bigcup_{Q<\infty} \mathcal{L}_n(H, Q)$. For a required *universal* all-word capability, define $Q_{\mathrm{on},\infty}$ as the infimum of budgets admitting one nonanticipative rule for every word at every finite horizon. A rule may use the current target symbol at the start of its interval, but not future symbols. Define $Q_{\mathrm{off},\infty}$ similarly when the entire requested word is supplied in advance. These are worst-case all-word budgets, not a total sum of realized heater powers over time. Here *online* means that the controller receives the requested letter s_k before selecting the held controls on interval $[(k-1)\tau, k\tau)$; no observation of $s_{k+1}, s_{k+2}, \ldots$ is available then. It does not mean that an unknown future physical disturbance is being rejected by output feedback. Each word has its own trajectory driven by the same causal lookup rule; distinct words are not claimed to occur simultaneously on one trajectory.

3 Sampled input operators and the one-piece obstruction

At the nominal parameters $\kappa_1 = \kappa_2 = 211/100$, $\tau = 1$, set

$$E = e^{-K/2}, \qquad C = \int_0^{1/2} e^{-Ks}\, ds = K^{-1}(I - E), \qquad A = E^2 = e^{-K}, \qquad D = \begin{bmatrix} EC|_P & C|_P \end{bmatrix}. \tag{4}$$

Here $D \in \mathbb{R}^{6\times 10}$ maps two held, five-port input phases to one sampled state increment: $x_k = Ax_{k-1} + Du_k$. In the one-piece case, $x_k = Ax_{k-1} + C_1 u_k$ with $C_1 = K^{-1}(I - A)|_P \in \mathbb{R}^{6\times 5}$.

Theorem 3.1 (Exact one-piece language collapse)**.** *At nominal parameters, for every finite $Q \geq 0$, $\mathcal{L}_1(1, Q) \subseteq \{22\}$ and $\mathcal{L}_n(1, Q) = \varnothing$ for every $n \geq 2$. Moreover, $\mathcal{L}_1(1, \infty) = \{22\}$; the target* 22 *is realized by a positive one-piece control of total power* 233.258410*.*

Proof. For each of the eight non-22 one-letter requests and each of the nine length-two requests $22t$, a nonnegative rational Farkas vector y certifies infeasibility. For a length-$m \leq 2$ request, stack the guard rows and write the sampled outputs as linear functions of the m nonnegative five-port

controls U. If $\mathcal{G}U \geq b$ denotes the resulting guard system, the corresponding vector in the certificate tables of Appendix D satisfies $y \geq 0$, $y^T b > 0$ and $y^T \mathcal{G} < 0$ componentwise; hence $y^T \mathcal{G} U \leq 0 < y^T b$ for all $U \geq 0$. Each strict sign is checked using outward rational enclosures for e^{-K}. The 17 vectors and their input-column sign summaries are printed in Appendix D, as exact decimal rationals. No power constraint is used in the contradiction.

The one-letter candidate 22 is achieved by a single control in vertex order,

$$u_{22}^{(1)} = (57.280683,\ 0,\ 76.951713,\ 0,\ 49.513007,\ 49.513007)^T,$$

which uses total available-port power 233.258410. Its nine guard margins are strictly positive under certified outward interval evaluation. Any longer admissible word would have a feasible length-two prefix; only 22 can be the first symbol, and each $22t$ is excluded. □

Remark 3.2. *The language assertion for $n = 1$ is a statement about the* union of budgets. *It does not claim that* 22 *is feasible for every small budget. The impossibility for $n \geq 2$, however, is independent of how large the finite phase-power budget is.*

4 The sharp two-piece budget

For the symmetric terminal symbol 22, exchange t_1 and t_2 and average any feasible two-piece input with its exchange. This preserves the constraints and cannot increase the maximum phase power. The reduced eight inputs are ordered

$$(u_r^E, u_w^E, u_p^E, u_t^E, u_r^L, u_w^L, u_p^L, u_t^L),$$

where each terminal receives u_t and therefore costs $2u_t$ in the corresponding phase. The active nonzero controls occupy reduced indices $\mathcal{I} = \{1, 4, 5, 7, 8\}$ (one-based). The active weak guards have normals

$$H_* = \begin{bmatrix} 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \end{bmatrix}, \quad b_* = (1, 1, 1, 4)^T. \tag{5}$$

Define $D_{\mathrm{sym}} \in \mathbb{R}^{6\times 8}$ by summing the two terminal columns of D within each phase (with other columns unaltered) and $G_* = H_* D_{\mathrm{sym}}$. Let $w = (1, 1, 1, 2, 1, 1, 1, 2)$ and w_E, w_L be its phase-separated restrictions to $\mathcal{I}$. The following 6×6 linear system fixes the active controls $v \in \mathbb{R}^5$ and common phase power q:

$$T \begin{pmatrix} v \\ q \end{pmatrix} = \begin{pmatrix} b_* \\ 0 \\ 0 \end{pmatrix}, \qquad T = \begin{bmatrix} (G_*)_{:,\mathcal{I}} & 0_{4\times 1} \\ w_E^T & -1 \\ w_L^T & -1 \end{bmatrix}. \tag{6}$$

In (6), $w_E, w_L \in \mathbb{R}^5$ have zero entries in inactive phases. Let $T_{6\leftarrow b}$ replace its last column by $b = (b_*^T, 0, 0)^T$.

Theorem 4.1 (One-sample analytic optimum)**.** *The smallest two-piece budget for the single initial symbol* 22 *equals*

$$Q_0 = \frac{\det(T_{6\leftarrow b})}{\det T}, \tag{7}$$

with the strictly certified bounds

$$201.131345777023810433521487478546245111831804926982562 7 < Q_0 < 201.131345777023810433564841709247362174559932503567920 9.$$

Proof. The unique solution of (6) has five strictly positive active controls and both phase budgets tight. Its four active guard equalities imply $r - p = 1$, $b - w = 1$, $p - t_i = 1$, and $t_i - b = 4$, hence $p - w = 6 > 5751/1000$. It is physically feasible. For the reduced dual, find $y \in \mathbb{R}^4_{\geq 0}$ and $(\lambda_E, \lambda_L) \in \mathbb{R}^2_{\geq 0}$ satisfying $\lambda_E + \lambda_L = 1$ and $(G_*)^T_{:,j} y \leq \lambda_{\mathrm{phase}(j)} w_j$, with equality for $j \in \mathcal{I}$. The associated six-by-six dual active matrix is nonsingular, and outward rational interval arithmetic proves all six dual variables positive and all three inactive reduced costs strictly positive. Weak duality makes the primal optimum global. Terminal-exchange averaging lifts the reduced result to all nonsymmetric feasible actions. Cramer's rule gives (7); the displayed enclosure is certified by a rational Taylor/uniformization bound and perturbation bounds for both six-by-six inverse systems. The active primal/dual construction and certified strict bounds are detailed in Appendix D. □

5 An exact causal all-word policy

Let $a = e^{-1}$, $\|A\|_\infty = a < 1/2$, and $A\mathbf{1} = a\mathbf{1}$. The controller specified by Appendix D contains initial actions u^0_s $(s \in \Sigma)$, bootstrap actions $u^B_{s,t}$ $(s, t \in \Sigma)$, and persistent actions $u^T_{s,t}$ $(s, t \in \Sigma)$. There are $9 + 81 + 81 = 171$ actions. The initial 22 action is the analytic optimizer of Theorem 4.1; the other 170 actions consist of nonnegative rationals on the 10^{-10} lattice. Their complete ten-port values are printed in Appendix D.

Let $M \in \mathbb{Q}$ be a common mean parameter, let $z^0_s \in \mathbb{Q}^6$ be nine initial reference states, and let $d_s \in \mathbb{Q}^6$ be nine persistent shape modes. All are explicitly given in Appendix D. The intended states are $z_{k,s} = (1 - a^k)M\mathbf{1} + d_s$ for $k \geq 2$. The policies satisfy the approximately affine identities

$$Az^0_s + Du^B_{s,t} = (1 - a^2)M\mathbf{1} + d_t + r^B_{s,t}, \tag{8}$$

$$Ad_s + Du^T_{s,t} = (1 - a)M\mathbf{1} + d_t + r^T_{s,t}, \tag{9}$$

with explicit uniformly small residuals. The exact plant runs with these finite lists; the ideal identities are only used for error certification.

Theorem 5.1 (Exact all-horizon causal power and zero preview gap)**.** *At the nominal parameters, for every $n \geq 1$, every word in Σ^n is realized by the above single nonanticipative two-piece policy with phase power at most Q_0, and*

$$\boxed{Q_{\mathrm{on},\infty} = Q_{\mathrm{off},\infty} = Q_0.}$$

Proof. Certified initial reference errors are at most δ_0; the bootstrap and persistent residuals in (8)–(9) are at most δ_B and δ_T in $\|\cdot\|_\infty$. Taking

$$R = 2(\delta_0 + \delta_B + \delta_T) < 4.425 \cdot 10^{-10},$$

we have $R \geq \delta_0$, $R \geq a\delta_0 + \delta_B$, and $R \geq aR + \delta_T$. Induction yields $\|x_k - z_{k,s_k}\|_\infty \leq R$ for every $k \geq 2$, with the correspondingly defined initial ideal state. Every guard is a difference of two coordinates, so a reference margin exceeding $2R$ remains positive in the physical state. The interval proof yields a uniform guard margin $> 7.84967 \cdot 10^{-5}$ for all first symbols other than 22 and all $k \geq 2$. The special initial 22 satisfies its four weak guards *exactly*, by the analytic active-basis equations; the stronger ownership guard remains strict. Every rational action has two phase sums strictly less than the *certified lower bound* on Q_0, while the analytic 22 action meets Q_0 exactly. Thus the policy realizes every word without preview at power Q_0.

Conversely any all-word policy, online or offline, must realize the one-letter word 22 and therefore needs at least Q_0 by Theorem 4.1. This proves the claimed equality. □

6 Strict and parameter-robust language separation

The exact optimal action for the first symbol 22 lies on a weak-guard boundary and is not itself a strictly robust witness. Let u^{210} be a second, strictly feasible initial 22 action with each held-phase power below 210. Its ten exact decimal-rational entries are printed in Appendix D. Let u^* denote the analytic optimum and $\Delta = D(u^{210} - u^*)$. Rational intervals show all six entries of Δ positive, $\operatorname{osc}(\Delta) < 401/1000$, and the u^{210} initial-22 guard margin exceeds $m_0 = 999/10000$. The exact causal policy of Theorem 5.1 has all other sampled-guard margins exceeding $G_0 = 7849/10^8$.

Lemma 6.1 (Oscillation contraction)**.** *If $A \geq 0$ and $A\mathbf{1} = a\mathbf{1}$ for $0 < a < 1$, then $\operatorname{osc}(A^j d) \leq a^j \operatorname{osc}(d)$ for all vectors d and $j \geq 1$. In particular $|(e_i - e_l)^T A^j d| \leq a^j \operatorname{osc}(d)$.*

Proof. Multiply $\min(d)\mathbf{1} \leq d \leq \max(d)\mathbf{1}$ by the nonnegative matrix A^j and use $A^j\mathbf{1} = a^j\mathbf{1}$. □

Replace only the special initial action by $u^\alpha = (1-\alpha)u^* + \alpha u^{210}$. The first 22 guard margin is greater than αm_0. Every later state differs from the original exact causal policy by $\alpha A^{k-1}\Delta$. By Lemma 6.1, $a < 3/8$ and $\operatorname{osc}(\Delta) < 401/1000$, each subsequent guard loses less than αH_0, where $H_0 = 151/1000$. The resulting nominal uniform margin therefore exceeds

$$\mu(\alpha) = \min\{\alpha m_0,\ G_0 - \alpha H_0\}, \quad m_0 = \frac{999}{10000}, \quad G_0 = \frac{7849}{10^8}, \quad H_0 = \frac{151}{1000}. \tag{10}$$

The mixed first action costs less than $Q_0 + 10\alpha$ in each phase.

Theorem 6.2 (Explicit parameter-robust phase and power tradeoff)**.** *For $0 < \varepsilon < 1/100$, choose α, η so that*

$$0 < \alpha < \min\{\varepsilon/10, G_0/H_0\}, \qquad 0 \leq \eta < \min\left\{10^{-8}, \frac{\alpha m_0}{24240}, \frac{G_0 - \alpha H_0}{24240}\right\}. \tag{11}$$

For every plant with $|\kappa_i - 211/100| \leq \eta$ and $|\tau - 1| \leq \eta$, the same finite-action, nonanticipative $H = 2$ policy realizes every word in Σ^n, for every n, within phase power $Q_0 + \varepsilon$ and with strictly satisfied sampled guards. In the same parameter box, $\mathcal{L}_n(1, Q) = \varnothing$ for all finite Q when $n \geq 2$, while $\mathcal{L}_1(1, \infty) = \{22\}$.

Proof. All phases have power less than $Q_0 + \varepsilon < 202$. Positive leakage and nonnegative controls give $\sup_t \|x(t)\|_\infty < 202$. Uniform Duhamel estimates for $\eta < 10^{-8}$ yield $\|A_\theta - A_0\|_\infty \leq 20\eta$ and $\|D_\theta - D_0\|_\infty \leq 10\eta$. Since $\|A_\theta\|_\infty < 1/2$, the sample-to-sample state error for identical actions is bounded by 12120η at every horizon, and any difference guard changes by less than 24240η. The inequalities in (11), together with (10), guarantee positive residual guard margin.

For $H = 1$, all 17 strict Farkas certificates are stable under the same parameter variation: their nominal maximum reduced cost is less than $-1.6286883 \cdot 10^{-6}$ and its drift is bounded by 49η, which cannot reach zero for $\eta < 10^{-8}$. A strictly feasible one-letter 22 witness has margin > 0.003425 and loses less than 1872η in the same box. Thus the one-piece language assertion persists. The common power cap is a rate cap, unchanged by variation in the sampling interval. □

Corollary 6.3 (Numerical rational robustness cube)**.** *Set $\alpha = 1/3200$, $\varepsilon = 1/250$ and $\eta = 1/800000000$. Then throughout*

$$|\kappa_1 - 2.11|, \quad |\kappa_2 - 2.11|, \quad |\tau - 1| \leq 1.25 \cdot 10^{-9},$$

$H = 2$ realizes every word online at power $Q_0 + 0.004$, with sampled guard margin $> 9.1875 \cdot 10^{-7}$; the one-piece collapse persists. This is a sufficient *cube, not a computation of the maximum physical tolerance or of the sharp perturbed power threshold.*

Audit of the perturbation constants

Write $\theta = (\kappa_1, \kappa_2, \tau)$ and $\eta = \|\theta - \theta_0\|_\infty$, interpreting the norm coordinatewise. The Laplacian has row-sum zero; hence $K(\kappa)\mathbf{1} = \mathbf{1}$, $-K(\kappa)$ is Metzler, and every $e^{-K(\kappa)s}$ is nonnegative with infinity norm exactly e^{-s}. Moreover $\|K(\kappa) - K(\kappa_0)\|_\infty \le 4\eta$ for changes of the two pendant conductances. Duhamel's formula yields

$$e^{-K's} - e^{-Ks} = \int_0^s e^{-K'(s-v)}(K - K')e^{-Kv}\,dv, \quad \|e^{-K's} - e^{-Ks}\|_\infty \le 4s\eta.$$

For $|\tau - 1| \le \eta < 10^{-8}$, $\|K'\|_\infty < 16$ and $h = \tau/2$, this estimate and the time-variation estimate imply the conservative bounds

$$\|e^{-K'h} - e^{-K/2}\|_\infty < 10\eta, \quad \|\textstyle\int_0^h e^{-K's}ds - \int_0^{1/2} e^{-Ks}ds\|_\infty < 2\eta.$$

Consequently the sampled two-half maps obey $\|D' - D\|_\infty < 10\eta$, $\|A' - A\|_\infty < 20\eta$; the corresponding one-piece maps $C_1' = \int_0^\tau e^{-K's}ds$ and $A'C_1'$ obey $\|C_1' - C_1\|_\infty < 4\eta$ and $\|A'C_1' - AC_1\|_\infty < 24\eta$.

Because $\|A'\|_\infty = e^{-\tau} < 1/2$ and the shared heater sum is below 202, the state error $e_k = x_k' - x_k$ obeys $\|e_k\|_\infty \le \frac{1}{2}\|e_{k-1}\|_\infty + (20+10)202\eta$, giving the horizon-uniform estimate $\|e_k\|_\infty < 12120\eta$. For one-piece infeasibility, each guard normal has ℓ_1 norm 2; each Farkas vector printed in Appendix D is nonnegative and has total multiplier mass no more than $1 + 10^{-12}$. Thus the perturbation of each reduced input coefficient is bounded by

$$2\max\{4\eta, 24\eta\}(1 + 10^{-12}) < 49\eta.$$

The certified strict negative coefficients and positive right sides retain their signs over the stated robustness region. This argument checks *all* perturbed parameter points, not only the eight cube vertices.

7 An unbounded family of physically coupled thermal modules

The six-node example is not confined to one isolated plant. This section gives an exact, all-budget transfer to connected networks of unbounded size, while stating precisely which observation language is preserved. The projection relies on classical diffusion averaging and synchronous invariant subspaces, which are extensively studied in network dynamics [18, 19, 20]. The new claim here is the application of this exact intertwining to the particular certified $H = 1/H = 2$ language phase and its sharp all-word power.

Fix an integer $m \ge 1$ and m identical six-node modules indexed by $a = 1, \dots, m$. Each local module has the weighted edges and leakage of (1), including its individually disabled b-heater. For each node type $v \in V$, choose *arbitrary symmetric nonnegative intermodule conductances* $\gamma_{ab}^v = \gamma_{ba}^v \ge 0$ for $a \ne b$, independently for the six node types. These edges connect (a, v) to (b, v); no cross-type intermodule edge is permitted. Let L_v be the resulting $m \times m$ weighted Laplacian. With e_v the six standard coordinate vectors, the full $6m$-node thermal plant is

$$K_m = I_m \otimes K_0 + \sum_{v \in V} L_v \otimes e_v e_v^{\mathsf{T}}, \quad \dot{X} = -K_m X + U, \quad X(0) = 0, \tag{12}$$

where $K_0 = I_6 + L(211/100, 211/100)$ at nominal parameters. At every held phase, $U^a \ge 0$, $U_b^a = 0$, and the *global*, not per-module, power satisfies $\sum_{a=1}^m \sum_{v \ne b} U_v^a \le Q$. When at least one of the intermodule graphs is connected, the resulting entire thermal graph is connected; the couplings may otherwise be arbitrary, including arbitrarily large.

Synchronous local observation contract. A letter $s_k \in \Sigma$ is required in *every* module a at sample time $k\tau$, with exactly the same affine guards $H_{s_k} X_k^a \ge b_{s_k}$. Its region-independent native-record projection is calculated on the *induced local six-node tree* G_a, at the query vertices (a, t_i). This

local-graph choice is part of the mathematical task. It does not assert that the full $6m$-node graph's single global EF–FG record has the same terminals, ownership, or elder order. Different target words for different modules are not part of this theorem. Write $\mathcal{L}_n^{(m)}(H,Q)$ for the common-word language in this network, using the same H-piece hold class as before.

Define the averaging and synchronous-embedding maps

$$\mathsf{R}_m = \frac{1}{m}(\mathbf{1}_m^{\mathsf{T}} \otimes I_6), \qquad \mathsf{S}_m = \mathbf{1}_m \otimes I_6. \tag{13}$$

Lemma 7.1 (Exact thermal mean and synchronized-copy intertwining)**.** *For all m and all the allowed intermodule conductances,*

$$\mathsf{R}_m\mathsf{S}_m = I_6, \qquad \mathsf{R}_m K_m = K_0\mathsf{R}_m, \qquad K_m\mathsf{S}_m = \mathsf{S}_m K_0. \tag{14}$$

Thus the mean $\bar{x} = \mathsf{R}_m X$ obeys $\dot{\bar{x}} = -K_0\bar{x} + \bar{u}$, $\bar{u} = \mathsf{R}_m U$, while for common copies $X = \mathsf{S}_m x$, $U = \mathsf{S}_m u$ the intermodule heat currents vanish identically.

Proof. For each symmetric graph Laplacian L_v, both $L_v\mathbf{1}_m = 0$ and $\mathbf{1}_m^{\mathsf{T}}L_v = 0$. Left- and right-multiplication of (12) by the Kronecker maps in (13) annihilates every coupling term. The $I_m \otimes K_0$ term gives the displayed identities. Multiplying the full differential equation and using uniqueness proves the dynamic assertions. The equal-modules subspace is invariant even when the six intermodule graphs are mutually different. □

Theorem 7.2 (Exact language and sharp power scaling on coupled $6m$-node graphs)**.** *For every $m \geq 1$, every allowed symmetric intermodule coupling, every $Q \geq 0$, every $n \geq 1$, and $H \in \{1,2\}$,*

$$\boxed{\mathcal{L}_n^{(m)}(H,Q) = \mathcal{L}_n^{(1)}(H,Q/m).} \tag{15}$$

The equality also transfers existence of a single nonanticipative controller for all words; both the causal and full-preview all-word power thresholds scale by m. In particular, at the nominal plant,

$$\mathcal{L}_1^{(m)}(1,\infty) = \{22\}, \qquad \mathcal{L}_n^{(m)}(1,Q) = \varnothing \quad (n \geq 2,\ Q < \infty), \tag{16}$$
$$\mathcal{L}_n^{(m)}(2,mQ_0) = \Sigma^n \quad (n \geq 1), \qquad \boxed{Q_{\mathrm{on},\infty}^{(m)} = Q_{\mathrm{off},\infty}^{(m)} = mQ_0}.$$

The broadcast implementation uses the same 171 control templates and no future-letter preview, with the prescribed ten-port action copied simultaneously to all m modules. The 171 templates are not a controller-size lower bound.

Proof. Suppose an admissible m-module input realizes a common word $s_1 \cdots s_n$; the module controls may be completely unequal and may depend on their individual measured states. Lemma 7.1 gives a physically valid base-plant control $\bar{u}(t) = m^{-1}\sum_a U^a(t)$ with the same H-piece hold resolution, nonnegative available ports, and phase sum at most Q/m. Since each guard $H_{s_k}X_k^a \geq b_{s_k}$ is affine with an identical right-hand side, averaging gives $H_{s_k}\bar{x}_k \geq b_{s_k}$. Hence the full-system word is also a base-plant word at Q/m; this proves the first inclusion in (15) for *arbitrary nonuniform* module inputs.

Conversely, given any base-plant realization with phase budget Q/m, copy that control vector to all modules. By the second intertwining identity and the common zero initial field, the unique full-network solution remains $X(t) = \mathsf{S}_m x(t)$, regardless of coupling strength. Every module then satisfies all local sampled guards. Its total phase input is m times the base input, hence at most Q, proving the other inclusion.

For nonanticipative controllers, averaging a deterministic common-word policy's chosen inputs for every observed target prefix defines a base policy on the same prefixes, since there are no exogenous unknown disturbances or hidden adversarial inputs in the model. Replication carries any base policy back to the module network. The same argument works with full preview. Apply Theorems 3.1 and 5.1 to obtain (16); the first-letter 22 lower bound is inherited by averaging even if the full-network designer allocates heater power nonuniformly across modules. □

Corollary 7.3 (Network-size-independent robustness for homogeneous cell perturbations)**.** *Let all modules share the same perturbed κ_1, κ_2, τ from Corollary 6.3, while the symmetric intermodule conductances remain arbitrary and nonnegative. Then for every m, the $H = 1$ language remains empty after one sample at every finite global power, whereas the strict broadcast $H = 2$ controller realizes all common words at global power $m(Q_0 + 0.004)$. The certified sampled local-guard margin $> 9.1875 \cdot 10^{-7}$ is independent of m and of intermodule coupling strength. This is a sufficient global power budget; the sharp* perturbed *base threshold is not supplied.*

Proof. The two intertwining identities hold verbatim with the common perturbed six-node matrix $K(\kappa)$ in place of K_0. Replicate the strict base policy of Theorem 5.1, so that each local state is identically the base trajectory. Averaging the altered plant establishes the robust one-piece no-go. The per-cell control and margin are exactly those of the base theorem. □

Corollary 7.4 (Time-varying undirected intermodule diffusion)**.** *The exact language identity* (15)*, the nominal mQ_0 threshold, and the homogeneous-cell robustness conclusion continue to hold if each γ_{ab}^{v} varies arbitrarily with time, provided the intermodule conductances remain nonnegative, symmetric for each time, and locally bounded and piecewise continuous. The broadcast controller need not know the coupling schedule.*

Proof. At each time t, every symmetric intermodule Laplacian $L_v(t)$ has both left and right constant-vector kernels. Hence (14) holds pointwise with $K_m(t)$, and uniqueness for the corresponding nonautonomous linear system gives the same averaging and synchronous-embedding trajectories. All phase budget and affine-guard arguments are unchanged. □

Remark 7.5 (Scope and non-priority of the general synchrony principle)**.** *The operator identities* (14) *are classical properties of balanced diffusive couplings and equitable/quotient dynamics [18, 19, 20]. We do not claim a new general synchronization result or an arbitrary-graph EF–FG theorem. The exact mQ_0 conclusion is for the structured, identical-module, synchronous-*local*-observation family. It need not hold for different target words per module, cross-type intermodule edges, heterogeneous within-module plants, or EF–FG computed intrinsically on the entire joined graph. Those cases require separate analysis.*

8 Nonidentical modules: a uniform robust language phase

The synchrony equalities of Section 7 cease to hold when the modules have different internal dynamics. Nevertheless, the strict one-piece Farkas signs and a positive-semigroup perturbation argument allow a uniform theorem without any upper bound on the number or strength of intermodule links.

Nonidentical/time-dependent local plant. Keep the locally evaluated, simultaneous, same-letter observation contract and unit sampling period of Section 7. For $a = 1, \dots, m$, put

$$\mathcal{K}_\eta(t) = I_m \otimes K_0 + C(t) + \Delta(t), \quad \Delta(t) = \bigoplus_{a=1}^{m} \sum_{i=1}^{2} d_i^{(a)}(t)(e_b - e_{t_i})(e_b - e_{t_i})^T, \quad |d_i^{(a)}(t)| \le \eta. \quad (17)$$

Here $C(t)$ is any matching-node-type, nonnegative symmetric diffusive intermodule Laplacian, locally bounded and piecewise continuous. The $d_i^{(a)}(t)$ may be different across modules and may vary with time; $0 \le \eta \le 3/200000000$ ensures every terminal conductance stays strictly positive. The common leak remains exactly one, the starting field is zero and each b-heater is still closed. The total instantaneous power limit Q applies across *all* modules. In particular the local and intermodule coupling schedules are not assumed known to the controller. All statements below concern local sampled EF–FG projected bands, not global EF–FG states of the joined graph.

Lemma 8.1 (Uniform positive-semigroup row perturbation)**.** *Write $P_\eta(T,t)$ and $P_0(T,t)$ for the state-transition operators generated by $-\mathcal{K}_\eta$ and by the same system with $\Delta = 0$. Both are entrywise nonnegative, with every row sum and every column sum equal to $e^{-(T-t)}$. Furthermore, $\|\Delta(t)\|_1 = \|\Delta(t)\|_\infty \le 4\eta$. For any $q \in \mathbb{R}^6$ and $\widehat{q} = (\mathbf{1}_m^T \otimes q^T)/m$,*

$$\|\widehat{q}(P_\eta(T,t) - P_0(T,t))\|_{\max} \le \frac{4\eta}{m}\|q\|_\infty (T-t)e^{-(T-t)}. \tag{18}$$

Here $\|\cdot\|_{\max}$ is the largest absolute row entry.

Proof. Each edge perturbation is a symmetric signed Laplacian; at the common separator b the absolute column and row sums are at most 4η, and elsewhere at most 2η. Both full drift matrices are symmetric M-matrices with all row and column sums equal to one. Their nonautonomous transition operators are therefore positive and doubly substochastic, with the asserted exact mass sums. The unperturbed intermodule network has the exact averaging intertwiner $R_m P_0(T,s) = e^{-K_0(T-s)}R_m$, even for changing $C(s)$. Apply the Duhamel formula $P_\eta - P_0 = -\int_t^T P_0(T,s)\Delta(s)P_\eta(s,t)\,ds$. Using the column-sum bound for the positive base 6×6 heat semigroup, the column 1-norm bound on Δ, and the column sums of $P_\eta(s,t)$ gives (18) after integration. No graph-size or coupling-norm factor occurs. □

Theorem 8.2 (Heterogeneous all-size H1/H2 persistence)**.** *For every $m \ge 1$, all intermodule couplings $C(t)$ above, and every family of local terminal conductances satisfying*

$$|\kappa_i^{(a)}(t) - 211/100| \le \frac{3}{200000000} = 1.5\cdot 10^{-8}, \qquad i = 1,2, \tag{19}$$

we have, independently of the total heater budget,

$$\mathcal{L}_1^{(m)}(H1,Q) \subseteq \{22\}, \qquad \mathcal{L}_n^{(m)}(H1,Q) = \varnothing \quad (n \ge 2).$$

The unbounded-budget first-letter union is exactly $\{22\}$; the broadcast initial H1 witness realizes 22 at global budget $m\cdot 233.258410$. Conversely, the same 171-action, nonanticipative H2 lookup policy as in the nominal plant, broadcasting the input to every module and using the strict initial-22 mixture of $\alpha = 1/3200$, realizes $\mathcal{L}_n^{(m)}(H2,Q) = \Sigma^n$ for all $n \ge 1$ at common global budget

$$Q = m(Q_0 + 1/250). \tag{20}$$

Every sampled local scalar guard has a certified strict margin

$$> \frac{3121875}{10^{11}} - 8\cdot 202\cdot \frac{3}{200000000} = 6.97875\cdot 10^{-6}. \tag{21}$$

The policy requires neither future letters nor knowledge of the local parameter deviations and intermodule coupling schedules.

Proof. For the H1 obstruction, apply each of the 17 tabulated one-step or two-step nonnegative Farkas vectors to the *mean* guard arrays, which are valid whenever all modules must realize the same word. Each dual has coefficient sum $s \le 1 + 10^{-12}$, positive right-hand side, and nominal input-column coefficient at most $-\gamma/m$, with $\gamma > 1.6286883\cdot 10^{-6}$. By Lemma 8.1, an H1 held-input coefficient changes, for a one-step constraint, by at most $2\eta s/m$; for a two-step constraint, the first and second input blocks change by at most $6\eta s/m$ and $2\eta s/m$, respectively. These inequalities integrate $4\eta(T-t)/m$ on the relevant unit periods. Consequently all input coefficients remain strictly negative because $6\eta(1+10^{-12}) < 9.0001\cdot 10^{-8} < \gamma$. The positive Farkas right-hand sides are unchanged. This excludes the eight non-22 first letters and all nine possible successors after 22, *without using any power bound.* The nominal strict one-step H1 22 witness has minimum guard margin > 0.00342526, heater sum 233.258410 per module, and its broadcast error is bounded by $4\eta\cdot 234$ per coordinate. Its relational guard loss is at most $8\eta\cdot 234$, so the first letter is still attainable.

For H2, let $z(t) = \mathbf{1}_m \otimes x^0(t)$ be the sampled-word trajectory of the strictly feasible nominal 171-action policy and let $X_\eta(t)$ be the heterogeneous trajectory under its broadcast controls. The difference satisfies a positive-system variation-of-constants formula with forcing $-\Delta(t)z(t)$. Since every local heater component is nonnegative and the phase sum is at most $Q_0 + 0.004 < 202$, positivity and unit leakage give $\sup_t \|z(t)\|_\infty < 202$. Thus $\sup_t \|X_\eta(t) - z(t)\|_\infty \le 4\eta \cdot 202$, regardless of m and $C(t)$. Each guard compares two coordinates, so its loss is at most $8\eta \cdot 202$. The previous exact nominal strictification certifies every local sampled guard > 0.00003121875. Subtracting the uniform loss gives (21) for every horizon. □

Theorem 8.3 (Two-sided heterogeneous power bounds)**.** *Let $Q^{(m)}_{\mathrm{on},\infty}(\eta)$ and $Q^{(m)}_{\mathrm{off},\infty}(\eta)$ denote the smallest (infimal) budgets for all local synchronous words of all lengths for any fixed admissible nonidentical plant. Then at $0 \le \eta \le 1.5 \cdot 10^{-8}$,*

$$\frac{mQ_0}{1 + 168\eta} \le Q^{(m)}_{\mathrm{off},\infty}(\eta) \le Q^{(m)}_{\mathrm{on},\infty}(\eta) \le m(Q_0 + 0.004). \tag{22}$$

Moreover, for every $0 < \eta \le 10^{-8}$ the upper bound improves to

$$Q^{(m)}_{\mathrm{on},\infty}(\eta) \le m(Q_0 + 210000\eta), \tag{23}$$

with a guard-strict broadcast policy obtained by mixing the first 22 action with $\alpha = 20000\eta$. At $\eta = 0$ the sharp identity $Q^{(m)}_{\mathrm{on},\infty} = Q^{(m)}_{\mathrm{off},\infty} = mQ_0$ is recovered. Therefore the normalized online and offline optimal powers converge to Q_0 uniformly in m and the allowable coupling functions as $\eta \downarrow 0$.

Proof. Any all-word controller must realize the first letter 22 in every module. Apply the exact six-node first-22 analytical dual from Section 4 to the terminal-symmetrized *mean* output at time one. Its four nonnegative guard multipliers sum to $S < 84$ and the two phase multipliers sum to one; weak duality bounds its nominal value by Q/m even when the module heater allocations are unequal. The averaged physical field differs from the nominal averaged field, for each dual guard row, by at most $2\eta Q/m$: this is the time integral over $0 \le t \le 1$ of the row estimate (18) against $\|U(t)\|_1 \le Q$. Since the physical averaged first-22 guards hold, the nominal dual yields $Q_0 \le (Q/m)(1 + 2\eta S) < (Q/m)(1 + 168\eta)$. This proves the first inequality in (22); the remaining ones follow from Theorem 8.2 and the trivial comparison of preview with causal control.

For the near-nominal upper bound, choose $\alpha = 20000\eta$ and phase budget excess $\varepsilon = 210000\eta$. The original strictification argument gives initial-22 margin at least 0.0999α and every other margin at least $0.00007849 - 0.151\alpha$, while power remains strictly below $Q_0 + \varepsilon$ (since $10\alpha < \varepsilon$). The heterogeneous plant loses at most 1616η in any guard, because $Q_0 + \varepsilon < 202$. Thus the first guard retains at least 382η and the others at least $0.00007849 - 4636\eta > 0$ whenever $0 < \eta \le 10^{-8}$. The same causal policy then realizes all words at the budget of (23). At $\eta = 0$, use the exact nominal boundary policy, not the mixed strictly interior one. □

Relation to established results. Perturbation bounds for nonidentical diffusively coupled networks are well established [16, 17]; our Duhamel and norm inequalities are not a claimed general invention. The special-model contribution is that *both* a budget-free Farkas language obstruction and a common all-word causal control survive independently of module count and intermodule coupling strength. The exact equality mQ_0 need not persist under heterogeneity; (22) is a certified interval, not an assertion of an exact nonidentical-network optimum. This model still uses local synchronous words, a fixed sampling clock, fixed leakage, matching-type symmetric couplings and sampled-only constraints.

9 Cross-type coupling and independently prescribed local words

The exact modular equality in Theorem 7.2 assumes the same word in every module and links only matching node types. We now remove both restrictions, at the price of a quantitative bound on

intermodule coupling strength. This is a *different language contract*: at each sampling period k, every module a independently receives a letter $s_k^{(a)} \in \Sigma$. The combined alphabet is Σ^m, with $(9^m)^n = 9^{mn}$ length-n word sequences. Local EF–FG projections are evaluated on the induced six-node trees, not on the full coupled graph's global intrinsic basin.

Write $K_{\mathrm{iso}}(t) = \bigoplus_{a=1}^m K_a(t)$ with terminal edge perturbations as in (17), $|\kappa_i^{(a)}(t) - 211/100| \le \eta$. Let $C_\times(t)$ be an arbitrary *symmetric nonnegative intermodule graph Laplacian*, now allowed to link *different* local vertex types. Assume it is locally bounded and piecewise continuous and define its maximum instantaneous weighted degree

$$\Gamma = \sup_{t\ge 0} \max_{(a,v)} \sum_{(b,w):\ b\neq a} c_{(a,v),(b,w)}(t). \tag{24}$$

The plant is $\dot X = -(K_{\mathrm{iso}}(t) + C_\times(t))X + u$, $X(0) = 0$, with $u \ge 0$, every b-port closed, and a *single global* peak-power constraint $\mathbf{1}^T u(t) \le Q = mq$. No bound on the number of modules, the number of cross edges, or their change frequency beyond local regularity is required. The maximum weighted degree Γ, however, must be small for the next theorem.

Lemma 9.1 (Cross-type positive-semigroup perturbation)**.** *Let $Z(t)$ solve the uncoupled, possibly nonidentical block-diagonal plant $K_{\mathrm{iso}}(t)$ under the* same *nonnegative control u as the coupled trajectory $X(t)$. Then for any input with total peak power at most Q,*

$$\sup_{t\ge 0} \|X(t) - Z(t)\|_1 \le 2\Gamma Q. \tag{25}$$

If, in addition, every module's own peak input sum is at most q, and Y is the reference trajectory of m independent nominal *copies driven by these same per-module controls, then*

$$\sup_{t\ge 0} \|X(t) - Y(t)\|_\infty \le (\Gamma + 4\eta)q. \tag{26}$$

Both bounds hold uniformly in m, over arbitrary target-word histories, and for time-varying link weights.

Proof. All drift matrices are symmetric M-matrices with row and column sums one. Therefore the nonautonomous evolutions are nonnegative and have every row and column sum $e^{-(t-s)}$. Nonnegative control and zero initial temperature imply $\|Z(t)\|_1 \le Q$ and $0 \le Y_i(t) \le q$. The cross Laplacian has induced column norm $\|C_\times(t)\|_1 \le 2\Gamma$. Duhamel's formula for $X - Z$, using the positive ℓ_1 contraction, yields $\|X(t) - Z(t)\|_1 \le \int_0^t e^{-(t-s)} 2\Gamma Q\, ds \le 2\Gamma Q$. For the second statement, variation of constants relative to the nominal reference gives forcing $-[C_\times(t) + \Delta(t)]Y(t)$. At each vertex, the cross term is a weighted sum of coordinate differences of Y, so its magnitude is at most Γq (all nominal coordinates lie in $[0, q]$); the signed local terminal perturbation has infinity norm at most 4η and hence contributes at most $4\eta q$. The positive full-network evolution has infinity norm $e^{-(t-s)}$, which integrates to (26). □

Theorem 9.2 (Independent-word phase separation under weak cross-type links)**.** *Let $m \ge 1$, let every local terminal conductance satisfy $\eta \le 3/200000000 = 1.5 \cdot 10^{-8}$, and allow all symmetric cross-type intermodule connections satisfying*

$$0 \le \Gamma \le 10^{-9}, \qquad Q = m(Q_0 + 1/250), \quad Q_0 + 1/250 < 202. \tag{27}$$

For the complete vector-word *language with independently prescribed module letters:*

1. *Under H2, one copy of the original 171-action nonanticipative policy in each cell realizes* every *vector word in $(\Sigma^m)^n$ at all finite horizons $n \ge 1$, using no future letters or knowledge of the changing links. Every sampled guard remains strictly greater than*

$$0.00003121875 - 2 \cdot 202(4\eta + \Gamma) \ge 0.00000657475. \tag{28}$$

2. *Under H1 and at the* same stated global budget Q*, no vector word of length two is realizable, hence no longer vector word is realizable. This holds even when the heater budget is allocated arbitrarily and unevenly among modules and the controller knows the future word.*

The H1 assertion is finite-budget*; unlike Theorems 3.1 and 8.2, it is not an all-budgets impossibility claim for arbitrary cross-type coupling.*

Proof. H2: the nominal strict initial-22 mixture with $\alpha = 1/3200$ and the other 170 tabulated actions implement each requested word using its own local phase inputs; all have per-module peak sum below $q = Q_0 + 1/250$. This is the same causal policy with independent, not broadcast, inputs. The six-node rational certificate guarantees all nine guards of each independent reference trajectory a margin $> g_* := 0.00003121875$. By Lemma 9.1, every local coordinate difference changes by at most $2(\Gamma + 4\eta)q$, which is strictly smaller than $2 \cdot 202(\Gamma + 4\eta)$. The right-hand side of (28) is positive. Thus the same controls work for every array of requested words, all m, and all n.

H1: for each module a, fix its requested two-letter prefix. If its first letter is not 22, take the tabulated one-sample Farkas certificate for that first letter; otherwise take the two-sample certificate for its $22 \to s_2^{(a)}$ prefix. Thus *every* module has one of the 17 certificates. Its multipliers are nonnegative, their sum is at most $s_* := 1 + 10^{-12}$, and its rational right-hand side is at least

$$\delta_* := \frac{26059}{16000000000} = 1.6286875 \cdot 10^{-6} > 0. \tag{29}$$

The nominal signed input coefficients are all strictly negative, and Theorem 8.2 certifies their persistence for an *isolated* nonidentical local plant with $\eta \le 1.5 \cdot 10^{-8}$, even for time-varying conductances. Hence each certificate functional applied to its uncoupled local temperature observations $Z^{(a)}$ is nonpositive, for arbitrary nonnegative H1 controls.

If the coupled plant were to realize the two-letter vector word, the weighted observed guard inequalities would yield at least $m\delta_*$ after summing the m local Farkas functionals. Each weighted guard row compares two coordinates; because the nonnegative multipliers in a cell sum to at most s_*, the resulting coefficient at any coordinate/time has absolute value at most s_*. Consequently Lemma 9.1, at times one and two, bounds the discrepancy between the summed functionals on X and Z by

$$s_*(\|X(1) - Z(1)\|_1 + \|X(2) - Z(2)\|_1) \le 4\Gamma Q s_* = 4\Gamma m q s_*.$$

But $4 \cdot 10^{-9} \cdot 202(1 + 10^{-12}) < 8.081 \cdot 10^{-7} < \delta_*$, a contradiction after dividing by m. Thus every length-two vector word is excluded at the specified total budget. □

Theorem 9.3 (Strong-coupling obstructions are topology- and word-dependent)**.** *Consider two nominal local cells under zero initial temperature, unit leak, and arbitrary time-varying nonnegative heaters with* total *instantaneous budget Q. The H1/H2 distinction is irrelevant to both lower bounds below.*

1. *Add only the two* cross-type *links (r_1, w_2) and (r_2, w_1) of conductance g. If both cells satisfy the protected sampling guards at even one positive sample time, then*

$$Q \ge \frac{13.502(5 + 2g)}{4}. \tag{30}$$

   At $g = 60$ this is $Q \ge 421.9375 > 2(Q_0 + 0.004)$, so even the first common symbol $(22, 22)$ is impossible at the previous power budget.
2. *Instead connect* every corresponding node type *between the two cells by an edge of conductance g. If the first requested terminal-*1 *band in cell one is* 0 *and in cell two is* 2*, then*

$$Q \ge \frac{3}{2}(1 + 2g). \tag{31}$$

   At $g = 200$ this gives $Q \ge 601.5 > 2(Q_0 + 0.004)$. Hence even matching-type links can prevent independently different requested words at a fixed budget, although they preserve exactly the common-word *language of Theorem 7.2.*

These are analytical necessary conditions for any admissible time-varying control, not merely infeasibility reports from a finite H2 linear program.

Proof. For the crossed-link case set $Y(t) = (x_{r_1} - x_{w_1}) + (x_{r_2} - x_{w_2})$. Directly subtracting the local r, w heat equations and adding the two crossed-edge flows yields

$$\dot{Y} = -(5+2g)Y + 3\sum_{a=1}^{2}(x_{w_a} - x_{p_a}) + \sum_{a=1}^{2}(u_{r_a} - u_{w_a}).$$

The total thermal mass $S(t) = \mathbf{1}^T X(t)$ obeys $\dot{S} = -S + \mathbf{1}^T u \le -S + Q$, so $S(t) \le Q$. Positivity gives $\sum(x_{w_a} - x_{p_a}) \le S(t) \le Q$, and nonnegative inputs imply $\sum(u_{r_a} - u_{w_a}) \le Q$. Starting from $Y(0) = 0$ we get $Y(t) \le 4Q/(5+2g)$. But the two protected observations require $x_{r_a} - x_{w_a} \ge 1 + 5751/1000 = 6.751$, hence $Y(t) \ge 13.502$, proving (30).

For the matching-type case let $Z = X^{(2)} - X^{(1)}$. All six cross-links contribute $-2gZ$, so $\dot{Z} = -(K_0 + 2gI_6)Z + (u^{(2)} - u^{(1)})$. The local positive semigroup has infinity norm $e^{-(1+2g)t}$. Since $\|u^{(2)} - u^{(1)}\|_\infty \le Q$, the variation-of-constants formula gives $\|Z(t)\|_\infty \le Q/(1+2g)$. The two different first terminal bands require $g_1^{(2)} - g_1^{(1)} \ge 4 - 1 = 3$, whereas $|g_1^{(2)} - g_1^{(1)}| = |Z_{t_1} - Z_b| \le 2\|Z\|_\infty$. This implies (31). □

Interpretation against prior work. Positive semigroup and input-to-state sensitivity bounds, small-gain arguments for compositional temporal contracts [27], and synchronization constraints in diffusive networks [28] are existing methods, not new general principles claimed here. The additional certified fact is a concrete *topology-dependent phase separation* for region-independent quantitative EF–FG local observations: weak arbitrary-type links permit independent words and preserve the two-step H1 obstruction at a finite power, whereas two explicit strong-coupling networks force a larger power already at the first observation. Neither a global EF–FG color theorem for the joined graph nor a CCR-specific advantage follows.

10 A certified first-sample cross-link resolution phase

The preceding weak-link theorem guarantees all horizons at a very small degree cap; its strong-link examples supply coarse necessary power bounds. We now fix two *nominal, identical* six-node cells and replace the degree-bound assumption by two precise symmetric cross-type edges:

$$(r_1, w_2), \quad (r_2, w_1), \qquad \text{each of conductance } g > 0.$$

The local observations are computed on the induced six-node trees as in Section 9. Let K_g be the resulting 12×12 matrix $I + L_g$, and let P_{10} select the ten open local heater ports. Put

$$C_g = K_g^{-1}(I - e^{-K_g})P_{10}, \tag{32}$$

$$E_{g,h} = e^{-K_g/2}, \quad C_{g,h} = K_g^{-1}(I - E_{g,h})P_{10}, \tag{33}$$

$$D_g = [\, E_{g,h}C_{g,h} \mid C_{g,h} \,]. \tag{34}$$

For a first vector letter $(s,t) \in \Sigma^2$, the 18×12 guard matrix $H_{s,t}$ is the direct sum of the two local nine-row systems, with right side $b_{s,t}$. Hence the exact first-sample constraints are $H_{s,t}C_g v \ge b_{s,t}$ for H1 and $H_{s,t}D_g(v^-, v^+) \ge b_{s,t}$ for H2, with nonnegative controls and one global power cap applied to each held phase.

Theorem 10.1 (The 81-letter first-sample resolution phase)**.** *For either of the* two exact *cross-conductances*

$$g = \frac{7}{1000} \qquad \textit{or} \qquad g = \frac{1}{100},$$

H1 has empty *first-sample vector language at every finite peak-power budget, while H2 realizes* all *81 first vector letters at global per-phase power* $Q = 421$*:*

$$\mathcal{L}_1^{(2)}(H1, Q) = \varnothing \quad (Q < \infty), \qquad \mathcal{L}_1^{(2)}(H2, 421) = \Sigma^2.$$

At $g = 7/1000$ *the 81 exact nonnegative Farkas vectors have a positive right-side pairing* $> 2.03 \cdot 10^{-6}$ *and every reduced input coefficient* $< -2.03 \cdot 10^{-6}$*. The 81 rational H2 actions have every sampled guard margin* > 0.1047*. At* $g = 1/100$ *the corresponding bounds are* $> 6.03 \cdot 10^{-6}$*,* $< -6.03 \cdot 10^{-6}$*, and* > 0.1039*, respectively.*

Proof. For each (s, t), the certificate file supplies $y_{s,t} \in \mathbb{Q}^{18}_{\geq 0}$ such that

$$y_{s,t}^{\mathsf{T}} b_{s,t} > 0, \qquad y_{s,t}^{\mathsf{T}} H_{s,t} C_g < 0 \quad \text{coordinatewise on all ten heater columns.}$$

If $v \geq 0$ satisfied the H1 constraints, multiplying by $y_{s,t}$ would give $0 < y_{s,t}^T b_{s,t} \leq y_{s,t}^T H_{s,t} C_g v \leq 0$, a contradiction. This proves the H1 statement *without* a power bound. For H2 the same file supplies a nonnegative rational pair $(v_{s,t}^-, v_{s,t}^+) \in \mathbb{Q}_+^{20}$ with each ten-port phase sum < 421 and all 18 sampled guards strictly satisfied. Each of the 81 dual/primal pairs is checked using rational upper/lower envelopes for the matrix exponentials by the outward-rational verification procedure in Appendix H; no floating-point solver output is used to certify a sign. □

Proposition 10.2 (A certified change between two coupling strengths)**.** *At* $g = 1/200 = 0.005$*, H1 realizes the common first vector letter* $(22, 22)$ *by applying the same full-interval five-port held heater to both modules:*

$$v^{(a)} = \big(57.31316857, 0, 76.95531716, 49.53599898, 49.53599898\big), \quad a = 1, 2.$$

The total power is exactly 466.68096738*, and the least sampled guard exceeds* 0.0002973*. In contrast, Theorem 10.1 excludes* every *first vector letter at* $g = 0.007$*, for all power budgets. Therefore feasible first-step H1 behavior changes between these two exhibited coupling values. This early two-point comparison alone does* not *establish a unique threshold; a subsequent local theorem sharpens it analytically.*

Proof. Under the symmetric broadcast input, both cell temperatures are identical, and the cross-links act exactly as an added local r–w conductance g. With the quotient drift $K_0 + g(e_r - e_w)(e_r - e_w)^T$, the displayed rational held input yields nine strict local sampled guards in both cells; the least is bounded as stated by the exact positive-series exponential envelope. The other endpoint is Theorem 10.1. □

Theorem 10.3 (Strong-link structural cone obstruction)**.** *For the same two crossed links but* $g = 5$*, H1 cannot satisfy even the five common structural guards in both modules at the* first *sample, under any finite nonnegative heater power. Nevertheless, H2 realizes the common first vector letter* $(22, 22)$ *with total per-phase power at most* 702 *and minimum strict guard* > 0.096*.*

Proof. For arbitrary, not necessarily identical, module controls, the averaged state $\bar{x} = (x^{(1)} + x^{(2)})/2$ obeys the *exact* quotient identity

$$\dot{\bar{x}} = -(K_0 + g(e_r - e_w)(e_r - e_w)^T)\bar{x} + \bar{u}, \qquad \bar{u} = (u^{(1)} + u^{(2)})/2 \geq 0. \tag{35}$$

If both module states obey the five common affine difference guards at the first sample, their average does too. At $g = 5$, the quotient's r–w conductance is 7. For these five guards take the nonnegative rational multiplier

$$\lambda = (0.287949535,\ 0,\ 0.473411111,\ 0.119319677,\ 0.119319677).$$

It has $\lambda^T b > 0$ and all five coefficients of $\lambda^T H_{\text{struct}}(K_0 + 5L_{rw})^{-1}(I - e^{-(K_0+5L_{rw})})P_5$ are strictly less than -0.0025, as certified with rational matrix-exponential enclosures. Thus no constant nonnegative

input can meet the averaged structural guards. For H2 use the same two five-port phase controls in both modules,

$$v^- = (0, 0, 0, 34.77637637, 315.22362363),$$
$$v^+ = (132.78157732, 0, 96.03257593, 59.22748101, 0).$$

The early/late sums are 350 and 288.04163426, respectively, per module. The quotient under this symmetric broadcast has all nine requested 22 guards strictly above 0.096, using the same rigorous exponential enclosure. The global phase-power cap 702 follows. □

Remark 10.4 (What has and has not changed)**.** *The two-cell language here is the vector of* locally evaluated *EF–FG-compatible numerical terminal records. Joining the graphs can change the global elder/merge geometry: for example, two equal-height local elder maxima become competing equal-height maxima in one connected graph, whose global elder/tie resolution is not two separate local elder resolutions. No full-global-EF–FG equivalence, all-horizon H2 claim at* $g = 0.007$ *or* 0.01*, or* global *monotone sharp threshold on all* $g > 0$ *follows. The exact quotient map* (35) *is a standard balanced-diffusion argument; the special-model certified language consequences are the claim here [19, 22, 11].*

11 An exact local cross-link critical threshold

The preceding theorem certifies two separated cross-link strengths. We now obtain a sharp *local* phase classification, including a uniquely defined analytic critical value. Set

$$g_- = \frac{54757560}{10^{10}}, \qquad g_+ = \frac{54757561}{10^{10}}, \qquad I_g = [g_-, g_+].$$

These are the decimal values 0.0054757560 and 0.0054757561. We continue to require two identical, six-node nominal modules with exactly the two crossed r–w links defined above, positive input ports other than b, and the nine *local* EF–FG-compatible sample guards per module. No assertion here concerns the full joined graph's global extrema records.

Three-variable exact quotient. Write $v = e_r - e_w$, $\widetilde{K}(g) = K_0 + gvv^{\mathsf{T}}$, $T(g) = \widetilde{K}(g)^{-1}(I - e^{-\widetilde{K}(g)})$, and $S = [\, e_r \;\; e_p \;\; e_{t_1} + e_{t_2} \,]$. Define the three active-guard rows and the terminal-gap row by

$$A(g) = \begin{bmatrix} (e_r - e_p)^{\mathsf{T}} \\ (e_b - e_w)^{\mathsf{T}} \\ (e_p - e_{t_1})^{\mathsf{T}} \end{bmatrix} T(g)S, \qquad c(g) = (e_{t_1} - e_b)^{\mathsf{T}} T(g)S, \tag{36}$$

$$G(g) = c(g)A(g)^{-1}\mathbf{1}_3, \qquad \mathcal{D}(g) = \det \begin{bmatrix} A(g) & \mathbf{1}_3 \\ c(g) & 9/2 \end{bmatrix}. \tag{37}$$

Here $\mathbf{1}_3 = (1, 1, 1)^{\mathsf{T}}$. The exact equation $\mathcal{D}(g) = 0$ is equivalent to $G(g) = 9/2$ wherever $A(g)$ is invertible. It is an analytic matrix-exponential equation, not a claim that the root is rational.

Theorem 11.1 (Complete local first-sample phase and unique critical root)**.** *There exists a* unique $g_* \in (g_-, g_+)$ *with* $\mathcal{D}(g_*) = 0$*. For every* $g \in I_g$ *the following first-sample vector-language assertions hold:*

$$\mathcal{L}_1^{(2)}(H1, 467; g) = \begin{cases} \{(22, 22)\}, & g_- \le g \le g_*, \\ \varnothing, & g_* < g \le g_+; \end{cases} \tag{38}$$

$$\mathcal{L}_1^{(2)}(H1, Q; g) = \varnothing \qquad (g_* < g \le g_+, \ \textit{all finite } Q), \tag{39}$$

$$\mathcal{L}_1^{(2)}(H2, 421; g) = \Sigma^2 \qquad (g_- \le g \le g_+). \tag{40}$$

A high-precision numerical estimate *is* $g_* \simeq 0.00547575602911761758$*; only the stated rational bracket, uniqueness in that bracket, and the language statements are claimed as rigorously certified. No uniqueness or monotonicity assertion is made for* all $g > 0$*, and the H2 claim here is for* one observation.

Proof. Symmetric reduction. The plant, the desired first vector letter $(22, 22)$, the control constraints, and the guard polyhedra are invariant under interchange of the two modules and interchange of their two terminals. Any feasible nonnegative H1 control for this common letter can therefore be averaged over these permutations. The averaged input uses the same $(u_r, u_w, u_p, u_t, u_t)$ in both modules. The module trajectory then obeys the exact six-node drift $\widetilde{K}(g)$.

Primal face. Exact rational-exponential enclosures give $\|A(g)^{-1}\|_\infty < 80$ on I_g. The candidate three-input control $u(g) = A(g)^{-1}\mathbf{1}_3$ has strictly positive coordinates in the $r, p, t_1 + t_2$ grouping. Set $u_w = u_b = 0$ and apply u_t at each terminal in both modules. By construction,

$$x_r - x_p = x_b - x_w = x_p - x_{t_i} = 1, \quad x_{t_i} - x_b = G(g), \quad x_p - x_w = G(g) + 2.$$

The finite interval certificate below gives $4 < G(g) < 5$ and $2(u_r + u_p + 2u_t) < 467$ throughout I_g. Consequently all guards for $(22, 22)$ hold whenever $G(g) \le 9/2$. At the critical point an upper terminal band is tight, which is permitted by our closed sampled-band convention.

Matching dual. For a symmetric held input, use four of the nine local guard rows: $r - p \ge 1$, $b - w \ge 1$, $p - t_1 \ge 1$, and $b - t_1 \ge -9/2$. With the same three controlled columns, set $z(g) = c(g)A(g)^{-1}$ and take the nonnegative multiplier $y = (z_1, z_2, z_3, 1)$. The exact interval check yields $z_i > 1/2$ throughout I_g; its reduced coefficients vanish on the r, p, t inputs, and the reduced coefficient on the remaining w input is $< -1/5$. Its right-side pairing is exactly

$$y^{\mathsf{T}} b = z(g)\mathbf{1}_3 - \frac{9}{2} = G(g) - \frac{9}{2}.$$

If $G(g) > 9/2$, the Farkas contradiction excludes $(22, 22)$ even at unbounded heater power. Symmetrization transfers the contradiction to arbitrary two-module controls for this common letter.

Unique root. For $C(g) = \int_0^1 e^{-t\widetilde{K}(g)}\, dt$, positivity, the row sum e^{-t}, and Duhamel differentiation show $\|C'(g)\|_\infty \le 1$ and $\|C''(g)\|_\infty < 4/3$. Thus $\|A'\|_\infty \le 2$, $\|A''\|_\infty < 3$, $\|c\|_1, \|c'\|_1 \le 2$, and $\|c''\|_1 < 3$. Twice differentiating $G = cA^{-1}\mathbf{1}_3$ with $\|A^{-1}\|_\infty < 80$ yields

$$|G''(g)| \le 3(80) + 14(80)^2 + 16(80)^3 < 9 \cdot 10^6.$$

The pure-rational endpoint evaluation, including an absolute G-error less than $2.1 \cdot 10^{-20}$, establishes

$$G(g_-) - 9/2 < -1.0009 \cdot 10^{-10}, \qquad G(g_+) - 9/2 > 2.4365 \cdot 10^{-10},$$
$$\frac{G(g_+) - G(g_-)}{g_+ - g_-} > 3.43744549.$$

Since $g_+ - g_- = 10^{-10}$, the curvature bound implies $G'(g) > 3.4365 > 0$ for the entire interval. Therefore $G = 9/2$ has exactly one root there, and $\mathcal{D} = \det A\,(9/2 - G)$ has the same unique root.

All other 80 first vector letters. At g_+ there are 80 nonnegative rational Farkas multipliers, one for each vector letter other than $(22, 22)$, whose right-side pairings are > 0.00051730 and whose every held-port reduced cost is < -0.00047918. Their total multiplier mass is < 2. For the twelve-node cross-link generator, the g-derivative of its H1 input map has infinity norm at most one; hence a reduced coefficient can drift by at most $4|g - g_+| < 4 \cdot 10^{-10}$. The same 80 duals thus exclude these letters *uniformly* on I_g. Together with the primal–dual equivalence for $(22, 22)$, this proves (38) and (39).

H2 throughout the bracket. At g_+ the 81 rational two-phase witnesses each respect the global power cap 421 and have a strict minimum first-sample guard margin > 0.09644. The continuous-time

Duhamel formula bounds the first-sample state change for these fixed inputs by $421|g - g_+|$ in infinity norm; each two-coordinate guard changes by at most $842|g - g_+|$. This is below $8.42 \cdot 10^{-8}$ throughout I_g, so all 81 witnesses remain valid on the whole interval. All numerical signs used here are verified with outward exact-rational matrix-exponential bounds and the rational certificate tables of Appendix M. □

Remark 11.2 (Scope of the critical value)**.** *The number g_* is a* unique local critical coupling *for the* first-sample *H1 language in the precise two-cross-link network. It is not a globally unique H2-power threshold, a globally unique switching value for arbitrary g, a sharp power budget for the connected network, or a result for a global EF–FG observation on the combined graph.*

12 Complete all-horizon language phase for the crossed network

The first-step classification above leaves open whether the crossed two-cell plant can sustain independently chosen vector letters indefinitely. Here the answer is exact on the critical interval $I_g = [g_-, g_+]$ of Theorem 11.1, and yields a smaller explicit controller than the original six-node staged policy. The following result uses the *same* two cross links, held half-period heater model, initially zero thermal field, and *local* source-native numeric EF–FG readouts.

For each $s \in \Sigma^2$, let H be the common list of 18 two-coordinate guard normals and let $b_s \in \mathbb{Q}^{18}$ be the corresponding band-dependent right-hand side, so $Hx \geq b_s$ is precisely the sampled observation predicate. The normals do not depend on s; only b_s does. At fixed g put

$$E_g = e^{-K_g}, \quad C_{g,h} = K_g^{-1}(I - e^{-K_g/2})P_{10}, \qquad D_g = \begin{bmatrix} e^{-K_g/2}C_{g,h} & C_{g,h} \end{bmatrix}.$$

The input in the first ten columns of D_g acts on the first half-period, and the second ten columns on the second half-period.

Theorem 12.1 (A memoryless complete vector-word controller)**.** *For every $g \in I_g$, a fixed library of* 81 *nonnegative rational two-phase input vectors $u_s \in \mathbb{Q}^{20}_{\geq 0}$, indexed by the* current *vector target letter $s \in \Sigma^2$, has the following property. Both phase-power sums are strictly below* 421 *for each u_s, and for every finite $n \geq 1$ and every independently specified vector word $s_1 \cdots s_n \in (\Sigma^2)^n$ the zero-initial-state trajectory under $u_{s_1}, \ldots, u_{s_n}$ satisfies*

$$Hx_k \geq b_{s_k} \quad (1 \leq k \leq n), \qquad \min_{k,h}((Hx_k - b_{s_k})_h) > 0.0293355.$$

In particular,

$$\boxed{\mathcal{L}_n^{(2)}(H2, 421; g) = (\Sigma^2)^n \qquad (n \geq 1).} \tag{41}$$

The control at period k is a function only of s_k, not of any earlier or later target, and requires neither a state observer nor model-dependent online optimization.

Proof. It suffices first to work at g_+. With the rational actions in the action table in Appendix M, define for each signed guard normal h and integer lag $j \geq 1$ the worst independent predecessor contribution

$$m_{j,h} = \min_{a \in \Sigma^2} hE_{g_+}^j D_{g_+} u_a, \qquad a_{s,h} = hD_{g_+} u_s.$$

For any vector word, positivity and linearity give the exact sample formula

$$x_k = \sum_{j=0}^{k-1} E_{g_+}^j D_{g_+} u_{s_{k-j}}, \tag{42}$$

so, for $k \leq 12$, each current guard is bounded below by $a_{s_k,h} + \sum_{j=1}^{k-1} m_{j,h} - b_{s_k,h}$. The rational inequalities prove, for *every* s, h and all $\ell = 0, \ldots, 11$,

$$a_{s,h} + \sum_{j=1}^{\ell} m_{j,h} - b_{s,h} > 0.0397542, \tag{43}$$

including verified outward matrix-exponential and rational-action errors.

For $k > 12$, the omitted lags $j \geq 12$ satisfy $|hE_{g_+}^j D_{g_+} u_a| \leq 2 \cdot 421 e^{-j}$: the heat semigroup is positive with infinity norm e^{-j} and $0 \leq D_{g_+} u_a \leq 421\mathbf{1}$. Since $e^{-1} < 3/8$, their entire potential negative contribution is bounded by the *rational* tail

$$T = 2(421)\frac{(3/8)^{12}}{1 - 3/8} = \frac{223736661}{21474836480} < 0.010419. \tag{44}$$

Using all eleven explicit lag minima and this tail, the rational bounds prove the more conservative uniform guard margin > 0.02933572 at g_+.

Finally, for any $g \in I_g$, or even measurable time-varying $g(t)$ with values in I_g, the two crossed edges yield $\|K_{g(t)} - K_{g_+}\|_\infty \leq 2|g(t) - g_+|$. All inputs have pointwise total power at most 421, hence all physical coordinates are in $[0, 421]$. Positivity and Duhamel give $\|x_{g(\cdot)}(t) - x_{g_+}(t)\|_\infty \leq 842 \sup |g(t) - g_+|$. Each guard compares two coordinates, so its variation is at most $1684(g_+ - g_-) = 1.684 \cdot 10^{-7}$. This leaves more than 0.0293355 guard margin at every sample, uniformly in horizon. □

Theorem 12.2 (Power-independent two-sample H1 extinction)**.** *For every $g \in I_g$ and every finite peak-power budget Q,*

$$\boxed{\mathcal{L}_n^{(2)}(H1, Q; g) = \varnothing \qquad (n \geq 2).} \tag{45}$$

Proof. By Theorem 11.1, the only possibly feasible first vector letter is $(22, 22)$. At g_-, fix this first letter. For each of the 81 second vector letters t, let H_{22} and H_t be the corresponding 18-row local guard systems. The two-sample map with nonnegative constant heater vectors $v_1, v_2 \in \mathbb{R}_{\geq 0}^{10}$ is

$$x_1 = C_g v_1, \quad x_2 = E_g C_g v_1 + C_g v_2, \qquad C_g = K_g^{-1}(I - E_g)P_{10}.$$

Appendix M supplies an explicit $y_t \in \mathbb{Q}_{\geq 0}^{36}$ with

$$y_t^T(b_{22}, b_t) > 4.4101 \cdot 10^{-6}, \qquad y_t^T \begin{bmatrix} H_{22}C_{g_-} & 0 \\ H_t E_{g_-} C_{g_-} & H_t C_{g_-} \end{bmatrix} < -4.4101 \cdot 10^{-6} \mathbf{1}_{20}^T.$$

The inequalities are strict for all twenty input columns and are certified with exact-rational matrix-exponential enclosures. Farkas' lemma forbids any nonnegative two-step control, without assuming a finite upper bound on its magnitude. Across I_g, Duhamel estimates change any reduced coefficient by less than $7 \cdot 10^{-10}$; the multipliers remain valid, with every reduced coefficient still $< -4.4094 \cdot 10^{-6}$. Thus no $(22, 22) \to t$ can occur. Other first letters were already impossible, so every length-two word is excluded; every longer word has an excluded length-two prefix. □

Corollary 12.3 (Complete local all-horizon H1/H2 phase)**.** *Let g_* denote the unique root of Theorem 11.1. For every $g \in I_g$ and every $n \geq 1$, H2 realizes all 81^n independently requested vector words at global peak power 421. For H1, no length $n \geq 2$ word is possible at* any *finite power. At the first observation, the only H1 word is $(22, 22)$ whenever $g \leq g_*$ and $Q \geq 467$; for $g > g_*$, H1 has no first word at any finite power. The sufficient value 467 is not asserted to be the sharp first-word H1 budget, and 421 is not asserted to be the sharp all-word H2 budget.*

Corollary 12.4 (Time-varying cross-link robustness)**.** *The same 81-action H2 policy realizes* every *vector word of every length if the symmetric conductance of each of the two crossed edges changes together in time, measurably, within $|g(t) - g_+| \leq 10^{-5}$, at total phase power 421. Its sampled-guard margin exceeds* 0.01249*. No corresponding time-varying H1 extinction is claimed.*

Proof. Repeat the Duhamel comparison in Theorem 12.1: $1684 \cdot 10^{-5} = 0.01684 < 0.02933572 - 0.01249$. □

Significance and limits. The nonanticipative controller of Theorem 12.1 uses only the current vector letter, rather than a previous-letter mode. Its existence is certified by a finite polyhedral min-plus inequality and the positive semigroup tail, not by a finite simulation. Positive-system bounds, input-hold reachability, and affine certificates are standard tools; the special-model statement here is the full and exact first-vs-two-hold language classification in a nontrivially cross-coupled thermal graph. The theorem does not extend to arbitrary cross-link strengths or to the single global EF–FG color/ownership output of the joined graph.

13 Reproducibility and comparisons

No code or executable program is released with this article. Instead, the governing matrices, exact rational witnesses, finite input tables and strictly bounded verification method are included in the paper itself. Appendix D prints the 17 one-cell H1 Farkas certificates, the 170 rational two-hold controls (with the final initial action defined analytically), the 18 reference states, and a strict initial action for robustness. Appendix M prints all 81 crossed-network memoryless actions, all 81 two-step H1 duals and the first-step primal–dual families underlying the crossing-threshold and illustrative stronger-coupling results. Appendix H states a language-independent rational-arithmetic verification procedure and the exact matrix-exponential enclosure needed to check these finite certificates. The arithmetic signs and the contraction/tail arguments, rather than a finite prefix enumeration, establish the all-horizon assertions.

A reader may implement the deterministic checks in any programming language supporting exact integers and rational arithmetic: (i) form the graph Laplacians and port-incidence matrices as specified in Section 2 (or directly from the displayed edge lists); (ii) enclose each exponential with the positive Taylor/uniformization expansion and rational remainder; (iii) reconstruct the held-input maps; (iv) test every printed primal control and Farkas multiplier using rational interval comparisons; (v) verify the 171-action reference recurrence and the 81-action finite-lag minimum; and (vi) apply the proved positive-semigroup contraction for all remaining horizons. This is an independently reconstructible *algorithmic specification*; it does not assert a unique software implementation or release any unpublished source code. No external experimental dataset is used in deriving or testing the results of this article. The public measured-building datasets mentioned in the Introduction are cited to establish application context only; the preceding separate exploratory studies are not used as evidence for the exact H1/H2 claims. Finite full-network simulations may provide additional checks but are not needed for the stated theorems.

Interpretation of the resource certificate. The determinant ratio Q_0 is an exact expression in matrix-exponential entries, rather than an assertion that the number is rational or has a short closed form in elementary rational constants. The first 22 action is optimal with some weak guards tight, so nominal optimality alone does not imply parameter robustness at exactly Q_0. The robustness cube requires extra power and retains only the stated word-language properties, not the perturbed exact optimum. The 171-action table is a certificate for existence, not a minimal controller-size theorem. It does not improve on classical finite-state implementation in a machine-independent resource metric. The feasibility claims are quantified over every prescribed finite word with an interval-initially-revealed target letter; no unknown disturbances, intermediate-time guards, or arbitrary temporal formula are included.

Theorem-level relation to previous synthesis results. The 2020/2021 regular-language papers [14, 13, 15, 12] already establish general finite-state and data-driven synthesis frameworks for regular-language specifications. Robust finite abstractions [24], finite-memory LTL synthesis [25], positive-network LP control [9], and continuous-time AP-observation synthesis [29] likewise provide methodological precedent. Our claimed added result is not a faster synthesis algorithm, a new language formalism, or continuous-time-between-samples satisfaction: it is the matching of 17

fixed-plant no-go certificates, a closed-form first-letter power optimum, one explicit all-word causal invariant policy at exactly that optimum, and a quantitative robustness family. This comparison identifies a distinct theorem statement, but does not establish universal bibliographic priority.

The six-node causal policy uses a finite phase marker and the previous/current target labels after its initial phase. The new crossed-network H2 policy instead uses only the current two-cell target letter. It is directly executable by a conventional finite-state controller. The 171 listed actions are a constructive upper bound on a representation of the policy, not a proven lower bound on the smallest action library, memory, runtime, or device energy. Nor do our results imply any CCR-specific computational advantage. The exact nominal online/offline equality does not extend automatically to perturbed parameters. No guarantee is given between sample times, in the conservative $\dot{x} = -Lx + u$ model, or for arbitrary graph topologies outside the explicitly lifted modular family.

14 Conclusion

For a region-choice-independent numeric projection of source-native terminal records in one explicitly specified positive heat network, increasing held-input temporal resolution from one to two pieces changes the complete achievable language from a single feasible first letter and no extensions to all finite nine-symbol words. The two-piece language has a sharp, analytically certified all-horizon power threshold and needs no future target preview. Rational witness data and a positive-contraction argument also provide a concrete open perturbation family at slightly larger power. Moreover, a mean-mode projection and synchronous embedding lift the exact word languages, causal and preview thresholds, and robustness guarantee to an arbitrarily large family of physically interconnected thermal modules. This is a structured, broadcast-observation construction; no claim is made for the entire native EF–FG state of the full joined graph. For weak cross-type links, a new aggregated-dual argument still forbids all two-sample independent-vector words at a fixed total power while a componentwise causal two-hold policy realizes every vector word. Strong cross-type links can destroy even the common first symbol at that power; strong matching-type links may instead destroy incompatible independent first symbols. A further exactly certified cross-link phase identifies a locally unique determinant-defined critical coupling. On its rigorously certified narrow interval, 81 exact two-sample Farkas multipliers prove H1 extinction for every word of length at least two and every finite budget; a new 81-action memoryless H2 policy realizes every independently targeted vector word at every horizon with power 421, with a quantitative time-varying-coupling tolerance. The critical point is the unique root of a three-control determinant equation only on that narrow interval. These are special-model consequences of classical positivity, Farkas, Duhamel and synchronization methods, and do not amount to a universal language theorem for arbitrary thermally coupled graphs.

A Integrated exact certificate proofs and construction data

Every item needed for the stated finite primal–dual checks appears within this single publication. The following appendices constitute part of the article, not a separate supplement. Intermediate rational values written in decimal notation are exact decimal fractions unless explicitly declared as approximate. The mathematical theorems are proved by inequalities with outward rational error estimates, rather than by floating-point solver tolerances.

Explicit strict initial (22) action for robustness

In the port order (r, w, p, t_1, t_2), the early and late halves of u^{210} are

$$u^{210,-} = (81.216809,\ 0.010000,\ 0.010000) \oplus (0.010000,\ 128.267448),$$
$$u^{210,+} = (49.399026,\ 0.029877,\ 76.369600) \oplus (55.401182,\ 28.314571).$$

Here $\oplus$ means concatenation of coordinate tuples. The entries are exact six-place decimal rationals, both phase sums are below 210, and the strict initial guard and oscillation bounds required by the robustness theorem follow from the rational enclosure method in Appendix H.

B Exact modular lift certificate

For each $m \geq 1$, take m identical local six-node thermal trees G_a and connect only matching node types (a, v) across modules by independent, arbitrarily weighted undirected thermal edges. The $m \times m$ coupling Laplacians L_v have zero row and column sums. These identities also hold pointwise if the symmetric conductances vary with time; local boundedness and piecewise continuity give the needed ODE uniqueness, so the language and power equalities are unchanged. In module-major order, write $K_m = I_m \otimes K_0 + \sum_v L_v \otimes e_v e_v^T$, and define $R_m = \frac{1}{m}(\mathbf{1}_m^T \otimes I_6)$ and $S_m = \mathbf{1}_m \otimes I_6$. Then exactly (no floating-point approximation)

$$R_m S_m = I_6, \qquad R_m K_m = K_0 R_m, \qquad K_m S_m = S_m K_0.$$

The first identity transports the nonuniform-input mean to the same six-node heat equation; the second transports a six-node control to a synchronous $6m$-node trajectory with all coupling heat flows identically zero. The aggregate power transforms as $Q \mapsto Q/m$ under averaging and $q \mapsto mq$ under replication. For each common sample letter s_k, the affine guards $H_{s_k} x_k^a \geq b_{s_k}$ are preserved by averaging and by replication. Consequently, for $H = 1, 2$ and all n, Q, the synchronous local-observation language satisfies $\mathcal{L}_n^{(m)}(H, Q) = \mathcal{L}_n^{(1)}(H, Q/m)$. The same proof transfers online/all-preview policies since there are no unknown disturbances. Thus $H = 1$ has no word of length at least two, while the $H = 2$ all-word nominal threshold is exactly mQ_0 and has zero preview gap.

The lifted theorem concerns the *same target word in every induced six-node local tree*; it does not assert the corresponding claim for the full joined graph's global intrinsic EF–FG labels or independently different requested words. The averaging/synchrony technique is not a new general control-theory tool. For illustration, the matrix intertwining may be checked over rational numbers on independently assembled 6-, 12-, 18-, 24-, and 42-node networks. Direct $6m$-dimensional matrix exponentials on coupled examples, including the eight corners of the stated parameter cube, provide independent finite diagnostics. They are secondary to the symbolic all-m proof.

C Uniform heterogeneous-network certificate

Plant. For m six-node cells, symmetric nonnegative and locally bounded intermodule conductances may change with time. In every cell and at every time, $|\kappa_i^{(a)} - 2.11| \leq \eta$ for $i = 1, 2$, while the other three local edges, common leakage and sampling clock are unchanged. Writing Δ for the direct sum of two signed edge-Laplacian perturbations per cell, its row and column absolute sums are both at most 4η. The full positive evolution operator retains both row and column sums $e^{-(T-t)}$.

Mean-observation Duhamel inequality. For an arbitrary signed single-cell observation row q and $\widehat{q} = (\mathbf{1}_m^T \otimes q^T)/m$,

$$\|\widehat{q}[P_\eta(T, t) - P_0(T, t)]\|_{\max} \leq (4\eta/m)\|q\|_\infty (T - t)e^{-(T-t)}.$$

For H1 Farkas vectors the sum of multipliers is $\leq 1 + 10^{-12}$, and the original worst reduced input coefficient is less than $-1.6286883 \times 10^{-6}$ (after factoring $1/m$). For two sampled times the perturbed input coefficient moves by at most $(6\eta/m)(1 + 10^{-12})$. At $\eta = 3/200000000$ it stays less than $-1.5386883 \times 10^{-6}/m$. The dual right sides stay strictly positive; the no-go remains independent of heater budget.

H2 broadcast action. The analytic initial-22 action is mixed with the tabulated strict Q210 action at coefficient 1/3200. The other 170 tabulated actions are unchanged. The nominal local guard

margin is at least 0.00003121875 and the per-cell phase budget is less than $Q_0 + 0.004 < 202$. The actual nonidentical state stays within $4\eta \cdot 202$ in ℓ_∞ of the nominal broadcast state, independent of time, network size and coupling weights. Every two-coordinate guard thus retains a margin greater than

$$0.00003121875 - 1616\eta \geq 0.00000697875 \quad (\eta \leq 1.5 \times 10^{-8}).$$

The original H1 initial-22 nominal strict witness has margin > 0.00342526, and its distributed perturbation loses at most 1872η.

Two-sided exact-dual power bounds. The certified one-letter first-22 active guard dual has four nonnegative multipliers of sum

$$S \approx 83.071254203673391146293502469 1659 < 84$$

and two nonnegative phase multipliers summing to one. For arbitrary uneven full-network H2 heat inputs with instantaneous total power Q, the nominal averaged first-22 dual value cannot exceed Q/m. Duhamel shifts each of its averaged guard outputs by at most $2\eta Q/m$; therefore

$$mQ_0/(1 + 168\eta) \leq Q^{(m)}_{\mathrm{off},\infty} \leq Q^{(m)}_{\mathrm{on},\infty} \leq m(Q_0 + 0.004).$$

For $0 < \eta \leq 10^{-8}$, the stricter mixed policy with $\alpha = 20000\eta$, $\varepsilon = 210000\eta$ has nominal first margin $\geq 1998\eta$, subsequent margin $\geq 0.00007849 - 3020\eta$, and heterogeneous guard loss $< 1616\eta$, proving the improved online upper bound $m(Q_0 + 210000\eta)$. For $\eta = 0$ use the exact nominal Q_0 policy.

Exact verification. The perturbation constants, original Farkas and dual margins, and strict inequalities can all be recalculated by rational arithmetic from the matrices and printed tables in this paper. Direct numerical trajectories are illustrative; they are not required to prove the dimension-independent bound.

D One-cell exact Farkas duals and causal input ledgers

The nine state guards, in the required row order, are

$$\begin{aligned} &h_1 = e_r - e_p, && h_2 = e_p - e_w, && h_3 = e_b - e_w, \\ &h_4 = e_p - e_{t_1}, && h_5 = e_p - e_{t_2}, \\ &h_6 = e_{t_1} - e_b, && h_7 = e_b - e_{t_1}, \\ &h_8 = e_{t_2} - e_b, && h_9 = e_b - e_{t_2}. \end{aligned}$$

The associated right sides are $(1, 5751/1000, 1, 1, 1, L_{s_1}, -U_{s_1}, L_{s_2}, -U_{s_2})$, where $(L_0, U_0) = (1/2, 1)$, $(L_1, U_1) = (2, 3)$, and $(L_2, U_2) = (4, 9/2)$. For a one-letter word the dual vector has 9 entries; for a two-letter word it has 18 entries, ordered first-sample then second-sample. The vector components not listed are zero. Each component below is given as an exact decimal rational in this paper.

For each word w, the following table reports the exact y and a conservative bound on $\max_j (y^T \mathcal{G}_w)_j < 0$. This proves infeasibility even with unbounded nonnegative inputs.

word	$y^T b$	max reduced cost	nonzero dual entries $(j : y_j)$
00	1.013e-01	-6.294e-03	1:0.121099994839, 3:0.429530748355, 7:0.224684628403, 9:0.224684628403
01	4.223e-03	-4.223e-03	1:0.132558612598, 2:0.050714387924, 3:0.404486529776, 7:0.206120234851, 9:0.206120234851

word	$y^T b$	max reduced cost	nonzero dual entries ($j : y_j$)
02	2.886e-03	-2.886e-03	1:0.139953368585, 3:0.471767020387, 5:0.083442639411, 7:0.194139805514, 9:0.110697166103
10	4.223e-03	-4.223e-03	1:0.132558612598, 2:0.050714387924, 3:0.404486529776, 7:0.206120234851, 9:0.206120234851
11	2.042e-03	-2.042e-03	1:0.144623334131, 2:0.104111314225, 3:0.378117596847, 7:0.186573877399, 9:0.186573877399
12	1.128e-03	-1.128e-03	1:0.149681213084, 3:0.493559824797, 5:0.126496844546, 7:0.178379481059, 9:0.051882636513
20	2.886e-03	-2.886e-03	1:0.139953368585, 3:0.471767020387, 4:0.083442639411, 7:0.110697166103, 9:0.194139805514
21	1.128e-03	-1.128e-03	1:0.149681213084, 3:0.493559824797, 4:0.126496844546, 7:0.051882636513, 9:0.178379481059
2200	2.003e-03	-2.003e-03	1:0.037195497542, 3:0.100890039158, 7:0.066931878723, 9:0.066931878723, 10:0.102438528711, 11:0.063164254858, 12:0.281527844756, 16:0.140460038764, 18:0.140460038764
2201	1.470e-03	-1.470e-03	1:0.026753512969, 3:0.060191000891, 7:0.047396243996, 9:0.047396243996, 10:0.119448554593, 11:0.090097098897, 12:0.306976587710, 16:0.150870378474, 18:0.150870378474
2202	1.174e-03	-1.174e-03	1:0.020944204009, 3:0.037548438453, 7:0.036527760416, 9:0.036527760416, 10:0.128911937255, 12:0.426215735280, 14:0.105080956718, 16:0.156662082085, 18:0.051581125368
2210	1.470e-03	-1.470e-03	1:0.026753512969, 3:0.060191000891, 7:0.047396243996, 9:0.047396243996, 10:0.119448554593, 11:0.090097098897, 12:0.306976587710, 16:0.150870378474, 18:0.150870378474
2211	8.530e-04	-8.530e-04	1:0.014639788939, 3:0.012976133410, 7:0.024732995818, 9:0.024732995818, 10:0.139181849335, 11:0.121341833820, 12:0.336499620127, 16:0.162947391366, 18:0.162947391366
2212	5.329e-04	-5.329e-04	1:0.011606245494, 4:0.003166024156, 7:0.013410312219, 9:0.016576336375, 10:0.145992289250, 12:0.478213219423, 14:0.134159716821, 16:0.165517786541, 18:0.031358069720

word	$y^T b$	max reduced cost	nonzero dual entries $(j : y_j)$
2220	1.174e-03	-1.174e-03	1:0.020944204009, 3:0.037548438453, 7:0.036527760416, 9:0.036527760416, 10:0.128911937255, 12:0.426215735280, 13:0.105080956718, 16:0.051581125368, 18:0.156662082085
2221	5.329e-04	-5.329e-04	1:0.011606245494, 4:0.003166024156, 7:0.013410312219, 9:0.016576336375, 10:0.145992289250, 12:0.478213219423, 13:0.134159716821, 16:0.031358069720, 18:0.165517786541
2222	1.629e-06	-1.629e-06	1:0.012650846879, 4:0.009764870472, 5:0.004585966142, 9:0.005178904330, 10:0.150892615262, 12:0.491181525832, 13:0.149106289719, 16:0.013766345822, 18:0.162872635541

E Analytic initial 22 action

Let $(v_1, v_2, v_3, v_4, v_5, Q_0)$ solve $T(v_1, v_2, v_3, v_4, v_5, Q_0)^T = (1, 1, 1, 4, 0, 0)^T$. The **exact** row I–22 is

$$(v_1, 0, 0, v_2, v_2 \mid v_3, 0, v_4, v_5, v_5),$$

where the vertical stroke separates early and late half-interval port vectors. Its two phase sums equal Q_0 exactly. There is no rounding of this row to a ten-decimal fixed grid; the policy uses these real analytic coefficients.

F Exact common mean and reference shapes

The common mean is $M = \frac{164347980223}{5000000000}$. The nine vectors z_s^0 and d_s are shown as integers in units of 10^{-10}. Coordinate order: (r, w, p, b, t_1, t_2).

type	sym	r	w	p	b	t1	t2
z0	00	285185195436	167541458990	237719053301	178737308005	188736523029	188736523029
z0	01	289761207936	164615670333	227262641380	182052012922	192051227946	212051227946
z0	02	245554608451	173525632236	235553823475	183526417212	193525632236	225553038499
z0	10	250041410954	172282582576	240040625978	182283367552	212282582576	192282582576
z0	11	243315227169	155421997946	233314442193	193113997783	223113212807	223113212807
z0	12	243643722154	173640582250	233642937178	183641367226	203642152202	223642152202
z0	20	248496710078	167965302083	238495925102	183495925102	228495140126	193495140126
z0	21	243643722154	173640582250	233642937178	183641367226	223642152202	203642152202
z0	22	241898764484	171898764484	231898764484	181898764484	221898764484	221898764484
d	00	76961557465	-38932581080	28771198459	-28931796104	-18932581080	-18932581080
d	01	80205117876	-41542694250	15968090726	-31541909274	-21542694250	-1542694250
d	02	49828526159	-37966003887	22036351041	-27965218911	-17966003887	12035566065
d	10	80335412275	-41568753130	15942031846	-31567968154	-1568753130	-21568753130
d	11	37926624313	-37086995537	20423789439	-27086210561	2913004463	2913004463
d	12	33335439382	-36667700522	23334654406	-26666915546	-6666130570	13333869430
d	20	50607131176	-38121724891	21880630037	-28120939915	11879845061	-18121724891
d	21	32327479837	-37675660067	22326694861	-27674875091	12325909885	-1626332847
d	22	34124974753	-35878165151	24124189777	-25877380175	14123404801	14123404801

G Exact held-input tables

E denotes first half-interval and L denotes second half-interval. Every row has ten port numbers in the order $(r, w, p, t_1, t_2 \mid r, w, p, t_1, t_2)$. Initial rows have label `I-ss`; second-sample bootstraps are `B-ss-tt`; recurrent rows are `T-ss-tt`. The special analytic `I-22` row is given in Section B and is omitted from the following integer table. Every other control is exactly the integer vector shown divided by 10^{10}; zeros are printed explicitly.

Early phase (values in units of 10^{-10})

ID	r	w	p	t_1	t_2
I-00	0	0	0	1685834602178	240884723729
I-01	0	0	0	0	1996235849540
I-02	0	0	1053140108347	0	898868703837
I-10	0	0	796180122390	0	1122198021806
I-11	0	0	0	2011312672794	0
I-12	126865321333	0	964378972332	0	880097171130
I-20	430209356296	0	0	0	1535229056332
I-21	126865321333	0	964378972332	0	880097171130
B-00-00	0	0	0	299526561492	1711786111302
B-00-01	0	0	131157482245	0	1801845652849
B-00-02	0	442716494935	0	1568596177859	0
B-00-10	3573357728	0	0	1904085068592	0
B-00-11	0	0	894475165670	0	1013183260649
B-00-12	424861334062	0	0	0	1586451338733
B-00-20	0	0	665192774425	0	1258182052170
B-00-21	0	646855856073	0	0	1260802570246
B-00-22	737302046995	0	0	0	1274010625799
B-01-00	0	0	0	291018425249	1687710258661
B-01-01	0	4631342199	0	0	1870443095236
B-01-02	154591214025	0	0	0	1720483223410
B-01-10	0	5533709403	0	1869540728032	0
B-01-11	239327029169	0	0	1635747408266	0
B-01-12	0	454624957685	0	0	1420449479750
B-01-20	0	0	665192774425	0	1225598063286
B-01-21	0	672581933492	0	0	1215608119057
B-01-22	737302046995	0	0	1241426636915	0
B-02-00	0	0	63851419511	244384672407	1666187318074
B-02-01	0	130352723701	0	1711868827422	79154581114
B-02-02	0	0	745458517972	0	1147243521193
B-02-10	0	128516619917	0	106476594164	1684986796591
B-02-11	282472456611	0	0	0	1608872198612
B-02-12	0	521440058700	0	0	1369904596522
B-02-20	201218690640	0	0	0	1690125964583
B-02-21	0	713670957089	0	1177673698134	0
B-02-22	780447474436	0	0	1214551427261	0
B-10-00	0	0	55125985934	237303594044	1697177546479

ID	r	w	p	t_1	t_2
`B-10-01`	0	124520251327	0	1698944364529	111964904082
`B-10-02`	0	297084957962	0	0	1606631728607
`B-10-10`	40822869868	0	0	1862893816700	0
`B-10-11`	5049549705	0	1014821232867	0	883845903997
`B-10-12`	79574815209	0	939021385153	0	885120486207
`B-10-20`	0	481493467734	0	1513488884921	0
`B-10-21`	0	709430622698	0	1196779120690	0
`B-10-22`	774551559136	0	0	1232819373907	0
`B-11-00`	0	0	91733479532	195641740646	1658109586886
`B-11-01`	0	148990347078	0	1679022878361	68036261282
`B-11-02`	12087870658	0	755451597719	0	1101988515744
`B-11-10`	7114895307	0	209614716978	1652798371838	0
`B-11-11`	298953374253	0	0	1570574609869	0
`B-11-12`	353197526249	0	0	0	1516330457873
`B-11-20`	0	337130077995	0	0	1532397906127
`B-11-21`	9461881546	724540651368	0	1135525451208	0
`B-11-22`	795381688272	0	5780743399	0	1172019798925
`B-12-00`	0	0	68063755859	1682793572333	215096612417
`B-12-01`	0	0	199221238104	0	1688423164805
`B-12-02`	0	0	749670854320	0	1134561715463
`B-12-10`	0	0	196474409559	18285796681	1671843910639
`B-12-11`	0	33937868730	984460518984	0	865834182069
`B-12-12`	88317073686	0	939021385153	856894110944	0
`B-12-20`	0	490141655243	0	0	1482863523807
`B-12-21`	0	718078810207	0	0	1166153759576
`B-12-22`	783293817613	0	0	1204592998645	0
`B-20-00`	0	0	63053791276	214620497070	1692065675589
`B-20-01`	0	0	194211273522	0	1697219152714
`B-20-02`	0	0	737018381522	0	1149385799794
`B-20-10`	0	0	191464444976	1156787330	1697768907899
`B-20-11`	0	436602798781	0	1449801382535	0
`B-20-12`	84931760329	0	939021385153	0	862451035834
`B-20-20`	0	310773043388	0	0	1575631137928
`B-20-21`	0	709748479708	7452288264	1169203413344	0
`B-20-22`	779908504256	0	0	0	1210149923535
`B-21-00`	0	45496876715	0	241844477818	1687429846611
`B-21-01`	0	0	199221238104	1671871386656	16551778150
`B-21-02`	0	0	749670854320	1134561715463	0
`B-21-10`	0	0	196474409559	1632902677	1688496804643
`B-21-11`	0	33937868730	984460518984	0	865834182069
`B-21-12`	0	525847911819	0	0	1358384657965
`B-21-20`	13035532322	0	713965094500	0	1157231942961
`B-21-21`	0	718078810207	0	1166153759576	0
`B-21-22`	783293817613	0	0	0	1204592998645
`B-22-00`	0	0	74121327501	210136405910	1665038863786
`B-22-01`	0	82193031858	0	0	1787334187375
`B-22-02`	0	316961973623	0	0	1552565245611
`B-22-10`	0	83095399062	0	103817484894	1682614335277

ID	r	w	p	t_1	t_2
`B-22-11`	17885006419	0	1014821232867	836820979948	0
`B-22-12`	0	532186647344	0	0	1337340571889
`B-22-20`	208158232053	0	0	0	1661368987180
`B-22-21`	467788032669	0	0	1401739186565	0
`B-22-22`	787387015850	0	0	0	1185794449858
`T-00-00`	0	0	0	299541395583	1711771277212
`T-00-01`	2990659833	0	0	0	1904667766487
`T-00-02`	0	0	681607098461	0	1247984203507
`T-00-10`	0	0	128410653700	1803552195364	0
`T-00-11`	0	0	967168134227	0	955846034273
`T-00-12`	0	454624957685	0	1453033468635	0
`T-00-20`	0	0	665192774425	0	1258182052170
`T-00-21`	245823996863	0	642391114032	1019443315425	0
`T-00-22`	737302046995	0	0	0	1274010625799
`T-01-00`	0	4105409156	6641151817	1714796838262	285769273559
`T-01-01`	0	0	143940364807	0	1788267145076
`T-01-02`	0	246257428859	0	1661400997461	0
`T-01-10`	0	0	29904595627	1877753830693	0
`T-01-11`	0	0	979951016788	0	942267526500
`T-01-12`	43993675824	0	949335223483	0	914329527012
`T-01-20`	0	0	677975656987	0	1244603544398
`T-01-21`	422131021611	0	0	0	1485527404708
`T-01-22`	738970419752	0	10313838330	1262028414713	0
`T-02-00`	0	0	57297531057	261555668241	1676743073220
`T-02-01`	0	59199831574	0	0	1848458594746
`T-02-02`	0	0	738904629519	1174970272175	0
`T-02-10`	0	0	185708184757	48091958501	1682446305530
`T-02-11`	0	1860988098	1021681609570	884115828651	0
`T-02-12`	328808436862	0	0	1578849989457	0
`T-02-20`	0	0	722490305482	0	1185168120838
`T-02-21`	278780235159	0	650916325014	0	977961866147
`T-02-22`	766385580749	9529666935	0	1235397425111	0
`T-10-00`	0	0	12613364316	283522353694	1714441683320
`T-10-01`	0	0	143770846561	1703519497642	84977519427
`T-10-02`	0	0	694220462777	0	1234635567727
`T-10-10`	0	0	141024018016	70058643954	1720144915629
`T-10-11`	0	0	979781498542	0	942497398493
`T-10-12`	0	107944982606	852785747249	946927696465	0
`T-10-20`	0	0	606886562673	0	1300771863647
`T-10-21`	0	653524378789	0	0	1254134047531
`T-10-22`	741608203217	0	0	1269704469577	0
`T-11-00`	0	0	80078273341	1680350817743	228607908147
`T-11-01`	0	0	211235755586	0	1699491705945
`T-11-02`	0	509144771973	0	1496842730820	0
`T-11-10`	53213942894	0	0	157493961666	1696950521760
`T-11-11`	14510632440	0	1025771953975	867375839905	0
`T-11-12`	89035897944	0	949972106261	868650422114	0
`T-11-20`	339004001261	0	0	0	1672308671534

ID	r	w	p	t_1	t_2
`T-11-21`	292534591739	0	653341835140	0	961781999440
`T-11-22`	786942632161	0	0	1224370040633	0
`T-12-00`	0	0	84901350365	231797353503	1670309546182
`T-12-01`	0	0	216058832610	1692639879741	0
`T-12-02`	0	12237132540	748201583350	1147219710430	0
`T-12-10`	0	0	213312004065	18333643763	1676012778492
`T-12-11`	0	656789859001	0	0	1354522813794
`T-12-12`	0	726305886511	0	0	1277621267254
`T-12-20`	18008507674	0	723443124987	1166206793659	0
`T-12-21`	667438304	728673559584	0	1178317428432	0
`T-12-22`	783961255917	10594749377	0	1216756667500	0
`T-20-00`	0	0	0	279516148827	1704551175130
`T-20-01`	0	0	187442016905	1682851442804	47353933206
`T-20-02`	0	292841616863	0	1614816809456	0
`T-20-10`	0	0	142332784300	55677197310	1709648444710
`T-20-11`	0	0	1023452668886	0	884205757434
`T-20-12`	328080582722	0	0	1579577843598	0
`T-20-20`	0	298233925644	0	1609424500675	0
`T-20-21`	0	710205048791	0	1202504581949	0
`T-20-22`	769502009431	0	8631436020	0	1233179227344
`T-21-00`	0	0	89928903348	1670767821698	225271238974
`T-21-01`	0	0	221086385593	1676498530024	10073510702
`T-21-02`	210754737433	0	0	1696903688887	0
`T-21-10`	59736881135	0	0	1847921545185	0
`T-21-11`	0	660913511320	0	0	1350399161474
`T-21-12`	0	729666529122	0	1273871628625	0
`T-21-20`	20189917776	0	725242383806	1162226124738	0
`T-21-21`	2031508189	730684810368	0	1174942107763	0
`T-21-22`	790448203068	0	11277289305	0	1209587180421
`T-22-00`	0	0	74119533738	1665039046316	210137535984
`T-22-01`	0	0	205277015984	0	1665709562355
`T-22-02`	0	10069259041	740662925886	1118793975125	0
`T-22-10`	53657114508	0	0	1815869045543	0
`T-22-11`	0	448182160775	0	0	1421343999276
`T-22-12`	0	719098835186	0	0	1245719463433
`T-22-20`	17127518484	0	713965094500	0	1138433547067
`T-22-21`	295907753644	0	642391114032	931227292376	0
`T-22-22`	787385803776	0	0	0	1185794602750

Late phase (values in units of 10^{-10})

ID	r	w	p	t_1	t_2
`I-00`	886197076577	0	808399510823	0	305158072777
`I-01`	907150110441	0	727833742294	359529871363	16798948697

ID	r	w	p	t_1	t_2
`I-02`	642294749975	0	680139192275	376218786809	312659943735
`I-10`	681621675133	0	741389272953	452075463069	136226261639
`I-11`	738168411852	0	765206094146	41585142566	466353024231
`I-12`	602315737156	0	669938058953	423036580458	316022296227
`I-20`	609962237989	0	792827541649	535369767141	73153126015
`I-21`	602315737156	0	669938058953	501889474462	237169402223
`B-00-00`	860880120834	0	789308851011	298254222453	0
`B-00-01`	867337508782	0	704837964488	362721316813	61043539778
`B-00-02`	679760502944	2121217079	705653510747	55675787172	505232176356
`B-00-10`	876562270809	13532753045	716643886506	40652053153	363921709281
`B-00-11`	637700803141	7989902658	655083594947	462255959683	248282412364
`B-00-12`	555190453729	40986764988	730944940796	438754909799	182566124986
`B-00-20`	706425694179	0	688369793884	495492463860	111492242243
`B-00-21`	578021131965	4230161833	700077283923	525130473690	203853621384
`B-00-22`	463065292907	0	731576240526	542864328451	273806810910
`B-01-00`	860662551887	0	789297241969	298483400442	0
`B-01-01`	876124770126	13654377353	716424756821	362395255721	42713512773
`B-01-02`	704743045564	49771440739	739263079790	383057140067	134477966635
`B-01-10`	876535624778	13204088128	716321702439	46396956841	358854300608
`B-01-11`	630312211692	67375054748	721313021966	125188406742	467123977647
`B-01-12`	611132888672	35340974606	714191936629	437647370994	212999501893
`B-01-20`	706208125232	0	688358184843	493924813808	113289070284
`B-01-21`	573569809469	0	697511615893	523762701604	208513523140
`B-01-22`	462847723960	0	731564631484	279120553355	537779763995
`B-02-00`	868785985981	0	786235369579	305901505296	0
`B-02-01`	868313049418	0	706751686420	0	418032925391
`B-02-02`	709766646106	0	683867715266	373243509488	243611506956
`B-02-10`	869174575624	0	706920504672	417849371468	0
`B-02-11`	629889486274	67375054748	721238051853	469109344306	123700735613
`B-02-12`	613015404283	31372609440	710367022271	436322965566	220234671234
`B-02-20`	706589435014	47400897589	738874922854	499650744736	18796672601
`B-02-21`	579686078628	261796667	696240760524	273526236949	461597800026
`B-02-22`	462424998542	0	731489661372	283264764546	534133248335
`B-10-00`	867766898493	0	786657624788	304792964344	0
`B-10-01`	867604050780	0	706950795345	0	417523020924
`B-10-02`	714719048512	32126603264	722528541998	379719798372	162218680648
`B-10-10`	876070740003	13532753045	716571395951	43525184160	361612599635
`B-10-11`	638174088788	0	646098785210	455090877604	271948921193
`B-10-12`	602701699267	0	670106578114	421531318025	316973077388
`B-10-20`	687713421149	0	704140168481	179619745814	384483912690
`B-10-21`	578715061039	0	696281803598	266774235282	468029457478
`B-10-22`	462573762101	0	731503749971	276679941918	540555218805
`B-11-00`	871263634370	0	785121214917	308857681580	0
`B-11-01`	869799811123	0	706350594096	0	419076163108
`B-11-02`	709849953381	0	683590388294	366157537334	251714793786
`B-11-10`	877075626192	0	701381949706	81327261561	351527835335
`B-11-11`	629362651298	67758844415	721585870880	130458311976	462146994226
`B-11-12`	594497950436	62726412372	739980248634	427744030313	186364031039

ID	r	w	p	t_1	t_2
`B-11-20`	717694193433	27761440454	720301585185	493733576123	51821877599
`B-11-21`	579534267634	0	695978526629	275498847690	460301030841
`B-11-22`	461944507883	0	731409383066	532738459812	285220322034
`B-12-00`	869270148284	0	786030907086	0	306445100465
`B-12-01`	875727536233	0	701560020563	356598400407	75357334195
`B-12-02`	710250808409	0	683663252772	371118691752	246279919860
`B-12-10`	876491444263	0	701799086852	431583725656	0
`B-12-11`	638873204688	0	646420932642	456195218124	269823317340
`B-12-12`	602473300078	0	670085075688	241225689575	497528607454
`B-12-20`	688756881961	0	703844324339	499917817269	64950339963
`B-12-21`	579758521850	0	695985959456	520186567152	215381624336
`B-12-22`	462345362912	0	731482247544	283302362512	534182699826
`B-20-00`	868633459180	0	786267371474	305866134930	0
`B-20-01`	875090847128	0	701796484951	352401969962	78974799105
`B-20-02`	710183240280	840011048	684602344743	366992511611	248694565112
`B-20-10`	875854755158	0	702035551240	431004760121	0
`B-20-11`	644907298582	41443801013	696728757739	157947778162	470285037298
`B-20-12`	602507167036	0	670087179200	417925667081	320792659477
`B-20-20`	717265894689	28943080404	721427723386	494281736083	49394238232
`B-20-21`	579564722560	0	695903711269	268887513788	466956725177
`B-20-22`	462379229870	0	731484351057	533432653457	284016438410
`B-21-00`	866851264268	0	787771585532	301775367186	0
`B-21-01`	875727536233	0	701560020563	0	431955734602
`B-21-02`	710250808409	0	683663252772	127994389092	489404222521
`B-21-10`	876491444263	0	701799086852	431583725656	0
`B-21-11`	638873204688	0	646420932642	452678303720	273340231744
`B-21-12`	613087847504	31110812773	710112221204	430754200105	226247591208
`B-21-20`	712233669598	0	685994283044	486124218062	126960502090
`B-21-21`	579758521850	0	695985959456	270390360199	465177831289
`B-21-22`	462345362912	0	731482247544	534182699826	283302362512
`B-22-00`	869935675423	0	785734472048	307259951882	0
`B-22-01`	878153181385	9047737495	711976216717	355374323399	56761213798
`B-22-02`	715031196738	30946042951	721378389719	376778404174	167178639212
`B-22-10`	878564036037	8597448271	711873162334	412278026152	0
`B-22-11`	637800444261	0	646064214580	275360153909	452087860045
`B-22-12`	613161299931	30734334748	709743396524	430626438672	227047202918
`B-22-20`	706364554046	47400897589	738854440823	493920900924	24771879413
`B-22-21`	554838267424	42649025341	736274491656	218263595401	459287292972
`B-22-22`	462200117574	0	731469179340	534035286842	283608089038
`T-00-00`	860885199960	0	789310037490	298247956848	0
`T-00-01`	876187628572	13929447760	716697482729	363945414735	40552698999
`T-00-02`	701865860085	0	686942383177	377238475356	231962992640
`T-00-10`	868106495938	0	705078217256	60674200672	362712381367
`T-00-11`	632292594906	0	648401624101	461584629615	259720095737
`T-00-12`	611355536745	35340974606	714204732150	132346647046	518064782248
`T-00-20`	706430773305	0	688370980364	495489331058	111489109441
`T-00-21`	560858582288	0	688809978687	300678621630	460965490189
`T-00-22`	463070372033	0	731577427005	542861195649	273803678108

ID	r	w	p	t_1	t_2
`T-01-00`	860863731075	0	789300928460	0	298278534762
`T-01-01`	867539386942	0	704672971876	363268798446	60941743681
`T-01-02`	713245662268	35830161406	726167992871	35214994456	500853861794
`T-01-10`	876590740480	12232101952	715143584814	46577190850	360769054698
`T-01-11`	632489393941	0	648235445010	462135244051	259621432442
`T-01-12`	603064640312	0	670129575350	427939582936	310178874196
`T-01-20`	706627572339	0	688204801272	496039945493	111390446146
`T-01-21`	555492166780	43333771974	737095858635	523822070585	151568804821
`T-01-22`	462936703146	0	731526747207	276919567533	539929654908
`T-02-00`	866902332033	0	786925138838	304148202054	0
`T-02-01`	877651009021	10979375873	713902899571	362788485731	45990902598
`T-02-02`	707882992158	0	684557484525	126975291853	488126421349
`T-02-10`	874123628012	0	702693318604	429286827245	0
`T-02-11`	638210785807	0	646087925543	272786866542	454227094903
`T-02-12`	595091038802	62908620477	740248155406	103025628612	510039229497
`T-02-20`	712447905378	0	685986081712	493367393957	119511291747
`T-02-21`	560347808947	0	688706597894	514538906125	247719359828
`T-02-22`	462820042562	0	731570542215	280647458083	536274629934
`T-10-00`	861058325000	0	789152778556	298678055429	0
`T-10-01`	867515712948	0	704681892033	0	424188689566
`T-10-02`	702038985125	0	686785124243	374285522369	235346044207
`T-10-10`	868279620978	0	704920958322	423816680620	0
`T-10-11`	632465719946	0	648244365167	458631676628	263103147304
`T-10-12`	606016149173	0	671222769066	225861435263	508212319293
`T-10-20`	711885125021	7794983760	694733838763	493188274228	103710451022
`T-10-21`	578041183805	4520023483	700418799344	522342141434	205990524728
`T-10-22`	462855268732	685926804	732292021465	271907219161	543572236632
`T-11-00`	869426263522	0	785935120927	0	306592688831
`T-11-01`	875883651471	0	701464234404	356082516573	76020806395
`T-11-02`	683337765499	0	703046986850	65442748911	499845571912
`T-11-10`	875768280672	14259782999	717314834590	403969774533	0
`T-11-11`	637959421573	0	646031259901	272070711587	455251279734
`T-11-12`	602487032052	0	670039052805	238241973778	500544614159
`T-11-20`	665961113193	26772069825	730858382127	498155497924	26696131229
`T-11-21`	560147305141	0	688668776344	510311736935	252184854374
`T-11-22`	462271302770	727029954	732247188610	278746599228	537320552233
`T-12-00`	870230326781	0	785638828781	307315416297	0
`T-12-01`	876687714729	0	701167942258	11249799	432814800635
`T-12-02`	710560388306	0	683739358536	129981843594	487031082358
`T-12-10`	877451622759	0	701407008547	432454041488	0
`T-12-11`	611894810760	7639577392	676248860243	471118848562	181541097340
`T-12-12`	582312414970	0	691366871783	436863759177	242379692097
`T-12-20`	712208811136	0	685946477407	244334668319	368822715931
`T-12-21`	580023215747	0	696045434690	272460771180	462783251178
`T-12-22`	462610056808	0	731541722778	281855859089	535305034119
`T-20-00`	870952249432	6186402372	792153062128	295676619771	0
`T-20-01`	873218250426	0	702507556827	0	429528251042
`T-20-02`	714497615889	32951679194	723377148417	37946442891	502539786403

ID	r	w	p	t_1	t_2
`T-20-10`	877136794606	4656179154	706641286061	422878412973	0
`T-20-11`	638168257424	0	646070029961	454266802887	272807582522
`T-20-12`	595098519240	62915672870	740256970492	97661449445	515380060748
`T-20-20`	716952775098	30260869367	722761322565	157227111076	384110594687
`T-20-21`	577667851936	0	696632251157	264862873628	469085985725
`T-20-22`	462567930738	0	731474994722	536214229684	281055517651
`T-21-00`	870613778105	0	785474786039	0	307726973087
`T-21-01`	877071166054	0	701003899517	0	433237607224
`T-21-02`	704076777833	50520151901	739955011882	18826857890	497933873288
`T-21-10`	875678434131	14281464207	717324209557	45083303218	358945261682
`T-21-11`	611974021483	7514115026	676137642487	467776969180	185040446121
`T-21-12`	582517194586	0	691331404860	164487788801	514822287790
`T-21-20`	712160173134	0	685933450873	241800659806	371418388981
`T-21-21`	580029544822	0	696052773785	269825174751	465405179437
`T-21-22`	462271866448	0	731421415373	535308959343	282310431631
`T-22-00`	869935440044	0	785734556950	0	307259751771
`T-22-01`	876392827992	0	701263670427	352848833018	79921552890
`T-22-02`	710380758592	0	683752145393	131308414566	485871354244
`T-22-10`	875697100182	13532753045	716536826312	50453312287	355092680968
`T-22-11`	645127699786	40756065215	696063704559	464768982481	164596220754
`T-22-12`	582400697860	0	691186863246	432078669220	247848838798
`T-22-20`	712088428967	0	685981215831	485976818374	127266209622
`T-22-21`	559988332536	0	688701732014	310483013566	452139594678
`T-22-22`	462200122281	0	731469180331	534035300138	283608070044

H Proof-verification protocol

For each tested plant generator $K = I + L$ and held duration $h \in \{1, 1/2\}$, form P by selecting only the five open heater columns in each six-node cell. The exact transition operators are $E_h = e^{-hK}$ and $C_h = K^{-1}(I - E_h)P$. Use a rational number $s \geq \max_i (hK)_{ii}$ and set $B = sI - hK \geq 0$ componentwise, so that

$$e^{-hK} = \frac{\sum_{k\geq 0} B^k/k!}{\sum_{k\geq 0} s^k/k!}.$$

For a fixed integer cutoff $N = 115$, replace the two series by their rational partial sums. For $z \in [0, N+2)$, an upper bound on the omitted positive tail is

$$R_N(z) = \frac{z^{N+1}}{(N+1)!\,[1 - z/(N+2)]}.$$

Because $B \geq 0$ and its row sums equal $\beta = s - h$, its tail has entrywise upper bound $R_N(\beta)$. Combining the numerator and denominator tails gives the fully rational entrywise enclosure

$$\left|e^{-hK} - \widehat{E}_h\right|_{ij} \leq \left(\frac{1}{S_N(s)} - \frac{1}{S_N(s) + R_N(s)}\right) S_N(\beta) + \frac{R_N(\beta)}{S_N(s)},$$

$$\widehat{E}_h = \frac{\sum_{k=0}^{N} B^k/k!}{S_N(s)}, \qquad S_N(z) = \sum_{k=0}^{N} \frac{z^k}{k!}.$$

For the 12-node crossed plant one may take $(s, h) = (9, 1)$ or $(5, 1/2)$; the resulting enclosure has error less than 10^{-45}. Matrix inversion is exact over $\mathbb{Q}$, and ordinary outward-rational interval propagation through all matrix products yields certified lower and upper entries of C_h, $E_h C_h$ and the two-hold input map. Other sample rates use any larger rational s with $s < N + 2$.

For every printed Farkas row y, form the relevant stacked guard matrix H, right-hand side b and held-input map $\mathcal{C}$ in the displayed row order. Test $y_i \geq 0$, the rational inequality $y^T b > 0$ and upper bounds on every component of $y^T H\mathcal{C}$ strictly below zero. For every printed H2 action, sum each five- or ten-port half-phase vector to check its budget, multiply the rational exponential enclosures by the action and test the specified guard rows with lower endpoints. The residual bounds and contraction estimates appearing in the main proofs then extend finite tests to all words and all times specified in those theorems. In the crossed-network 81-action certificate, compute $m_{h,j} = \min_s hE^j Du_s$ for $1 \leq j \leq 11$, then apply the displayed geometric bound for all $j \geq 12$. All such operations involve rational integers, comparisons, additions, multiplications and exact linear algebra; no released software or external table is needed.

I Arbitrary-type intermodule couplings

Independent word alphabet. For m cells, each sample letter is the vector $(s^{(1)}, \ldots, s^{(m)}) \in \Sigma^m$. The cross-module links may join any different node types but have symmetric nonnegative conductances with weighted degree at most Γ at each vertex, even when they vary with time. Each module has its own five open actuator ports, the same closed b port, and the system's peak heater sum is at most $Q = mq$. The local terminal conductance deviations obey $|\kappa_i^{(a)}(t) - 2.11| \leq \eta$.

Exact constant audit. The prior fixed strict initial-22 mixture with $\alpha = 1/3200$ has nominal all-guard margin $g_* = 0.00003121875$, and the exact nominal power satisfies $q = Q_0 + 0.004 < 202$. Choose

$$\eta = \frac{3}{200000000} = 1.5 \cdot 10^{-8}, \qquad \Gamma = \frac{1}{10^9}, \qquad q < 202.$$

By leaky positivity, independent nominal state coordinates under each per-module H2 control lie in $[0, q]$. The time-varying total cross-Laplacian has at most Γ total conductance incident to each

vertex. Its action on the nominal state is bounded coordinatewise by Γq; signed terminal edge perturbations add at most $4\eta q$. The positive coupled semigroup has row sum e^{-t}, so the state infinity-norm deviation is at most $(\Gamma + 4\eta)q$. Each guard thus remains strictly larger than

$$g_* - 2(\Gamma + 4\eta)q > 0.00003121875 - 404(6.1 \cdot 10^{-8}) = \boxed{0.00000657475}.$$

The 171-action lookup is applied separately to each module's word; the resulting physical controls meet the global power cap and require no future labels.

Finite-power H1 obstruction for independent words. Fix a requested two-sample word in every module. Every module has a certified nonnegative Farkas multiplier vector: choose its first-symbol no-go if the first symbol is not 22, or the $22 \to s$ two-symbol no-go otherwise. All 17 witnesses have $\sum y_j \le 1 + 10^{-12}$ and $y^T c \ge \delta_* = 26059/16000000000$. The isolated nonidentical-cell H1 reduced input coefficients remain negative for $\eta \le 1.5 \cdot 10^{-8}$ by the earlier case. For the same input distribution, the coupled and isolated trajectories differ in ℓ_1 by at most $2\Gamma Q$ at every instant, since $\|C_\times\|_1 \le 2\Gamma$ and the isolated total heat is at most Q. Summing the cellwise duals over the first two observations, their entire cumulative perturbation is bounded by $4\Gamma Q(1 + 10^{-12})$. At $Q = mq$ and $q < 202$ this becomes per module

$$4\Gamma q(1 + 10^{-12}) < 8.081 \cdot 10^{-7} < 1.6286875 \cdot 10^{-6} \le \delta_*.$$

Hence no length-two vector word is feasible under H1 at this *fixed power budget*. The claim is not the prior all-budget no-go, which was proved only with matching-type module links.

Two strong-link lower certificates (without linear programming). For the two edges (r_1, w_2) and (r_2, w_1) of weight g, direct subtraction of the local r, w equations gives

$$Y' = -(5 + 2g)Y + 3\sum_a (w_a - p_a) + \sum_a (u_{r_a} - u_{w_a}), \quad Y = \sum_a (r_a - w_a).$$

Nonnegative heat and $\sum_i x_i(t) \le Q$ imply $Y' \le -(5 + 2g)Y + 4Q$. At any valid sample $Y \ge 13.502$, so $Q \ge 13.502(5 + 2g)/4$; at $g = 60$, this is exactly 421.9375. Even the first common $(22, 22)$ symbol fails at $Q = 2q < 404$.

For six corresponding-type edges of weight g between the two cells, the cell contrast $z = x^{(2)} - x^{(1)}$ obeys $z' = -(K_0 + 2gI)z + (u^{(2)} - u^{(1)})$. Thus $\|z\|_\infty \le Q/(1 + 2g)$, but the first symbols $(00, 22)$ require $g_1^{(2)} - g_1^{(1)} \ge 3$, giving $Q \ge 3(1 + 2g)/2$; at $g = 200$ this is 601.5.

Numerical cross-check. The adjacent Direct full-network matrix-exponential reconstruction independently recomputes complete matrix exponentials under different words, random cross-type edges and changing local parameters on 12, 18 and 30-node networks. It also solves a separate one-sample H2 linear feasibility problem for the two strong-link examples, both infeasible at the old two-module budget. The all-size/all-word guarantee rests on the exact inequalities and tabulated rational witnesses, *not* on these finite tests.

J Cross-link first-sample phase and quotient obstruction

Physical and logical contract. Two nominal six-node cells are joined by exactly the symmetric cross edges (r_1, w_2) and (r_2, w_1), each with weight g. Both separator heater ports remain shut, all other heater phases are nonnegative, and local terminal numerics are sampled on the induced six-node trees. The two-vector alphabet contains $9^2 = 81$ possible first letters; each supplies 18 affine guards. The claim for H2 in this section is *one sample only*, not all word lengths.

Machine-readable complete certificates. At $g = 7/1000$ Appendix M contains exactly 81 rows, indexed by $(s, t) \in \Sigma^2$. At $g = 1/100$ the same appendix contains the corresponding 81-row family. Each row has an H1 rational nonnegative 18-vector y, a 10-vector held-input reduced-cost target, and two H2 rational nonnegative ten-port phase actions. The rational-enclosure inequalities yield the following conservative strict bounds:

g	$\min y^T b$	$\max_j (y^T H C_g)_j$	H2 least guard
7/1000	$> 2.03 \cdot 10^{-6}$	$< -2.03 \cdot 10^{-6}$	> 0.1047
1/100	$> 6.03 \cdot 10^{-6}$	$< -6.03 \cdot 10^{-6}$	> 0.1039

In both cases the sum of all ten open heater inputs is strictly < 421 in each H2 half interval. H1 has no applicable power cap in the contradiction. The complete rational dual and action records for both conductances are printed in Appendix M.

Rigorous exponential interval algorithm. For either $A = K_g$ or $A = K_g/2$, select $s = 9$ and $s = 5$, respectively, so $B = sI - A \geq 0$ entrywise and all row sums are $\beta = s - 1$ or $s - 1/2$. Let $N = 115$. Construct the exact rational matrix polynomial $P_N(B) = \sum_{k=0}^{N} B^k/k!$ using integer matrix products with a common denominator. For any $z \geq 0$, the scalar Taylor remainder obeys

$$R_N(z) = \sum_{k=N+1}^{\infty} \frac{z^k}{k!} \leq \frac{z^{N+1}}{(N+1)!} \frac{1}{1 - z/(N+2)}.$$

Set $S_N(s) = \sum_{k=0}^{N} s^k/k!$ and $\widehat{E} = P_N(B)/S_N(s)$. Positivity gives the *entrywise*, rigorous error bound

$$|e^{-A} - \widehat{E}|_{ij} \leq \varepsilon_A := \left(\frac{1}{S_N(s)} - \frac{1}{S_N(s) + R_N(s)} \right) S_N(\beta) + \frac{R_N(\beta)}{S_N(s)}.$$

The rational enclosure evaluation yields $\varepsilon_A < 10^{-45}$ in both cases (actual bounds are much smaller). The inverse K_g^{-1} is computed as an exact rational nonnegative matrix with row sums one. Therefore the coefficient matrix $C_g = K_g^{-1}(I - e^{-K_g})P_{10}$ is within ε_{K_g} per entry of its rational approximation. The H1 signed guard multipliers add at most $2\varepsilon_{K_g} \sum_j y_j$ to any reduced coefficient. For H2, the early response $E_{g,h} K_g^{-1}(I - E_{g,h})P_{10}$ has entrywise error at most $2\varepsilon_{K_g/2}$, and the late response at most $\varepsilon_{K_g/2}$. Thus the loss in any two-coordinate guard under one control pair is bounded by $2\varepsilon_{K_g/2}(2\sum v^- + \sum v^+)$. All inequalities are checked with exact `Fraction` against these upper bounds; solver tolerances are not part of the proof.

A certified nonmonotonicity-free bracket. At $g = 1/200$ both modules can take the same rational full-period action

$$(57.31316857, 0, 76.95531716, 49.53599898, 49.53599898).$$

The full two-cell first letter $(22, 22)$ then has 9×2 guard margins greater than 0.0002973 under H1, and the global power is 466.68096738. The fraction proof uses $N = 115$ with the local mean quotient. At $g = 7/1000$, all 81 first letters are ruled out. This is an explicit change of feasible behavior between the *two tested conductances*; no uniqueness of a critical g or global monotonicity is inferred.

Large-link quotient obstruction. For $g = 5$ and arbitrary controls, the two-cell average $\bar{x}$ obeys

$$\dot{\bar{x}} = -(K_0 + 5(e_r - e_w)(e_r - e_w)^T)\bar{x} + \bar{u}.$$

The effective single-cell r–w conductance is 7. The common structural five-guard multiplier is

$$\lambda = (0.287949535, 0, 0.473411111, 0.119319677, 0.119319677).$$

Its rational-exponential reduced costs on the five allowed input ports are all < -0.0025 and its guard right-side pairing is > 0. Hence even the common structural chamber is H1-impossible at the first sample for unlimited nonnegative held inputs. For H2, replicate in both cells the rational phases

$$v^- = (0, 0, 0, 34.77637637, 315.22362363),$$
$$v^+ = (132.78157732, 0, 96.03257593, 59.22748101, 0).$$

The global per-phase power is at most 702, and each local 22 guard has a rigorous margin > 0.096.

Reproduction. All the displayed coefficient and guard inequalities follow from exact rational matrix enclosures and the certificate tables printed in this article. Floating-point candidate discovery, finite simulations and LP tolerances are not part of these sign proofs.

K Certified unique local coupling threshold

Exact analytic contract. On the interval $[g_-, g_+] = [0.0054757560, 0.0054757561]$, the two-module plant has exactly two cross-edges $(r_1, w_2), (r_2, w_1)$ of common weight g. The quotient matrices $T(g), A(g), c(g)$ and $G(g) = c(g)A(g)^{-1}\mathbf{1}_3$ are defined in the main text. The critical equation is

$$\det \begin{bmatrix} A(g) & \mathbf{1}_3 \\ c(g) & 9/2 \end{bmatrix} = 0.$$

The symmetrized common-first-letter candidate uses $u_w = u_b = 0$ and three independent nonnegative heater controls u_r, u_p, u_t, with u_t applied at both terminal ports. The active rows are $r - p$, $b - w$, and $p - t_i$, all equal to one. The remaining upper terminal guard is $t_i - b \le 9/2$. Its value is exactly $G(g)$.

Endpoint and curvature certificate. The pure-rational The exact endpoint certificate recomputes $e^{-\widetilde{K}(g)}$ via a positive Taylor-uniformization series, with a rigorously bounded tail, at both exact rational endpoints. After rounding the resulting exact rational approximations to a 10^{-25} grid and allowing the rounding error, the full matrix T has an entrywise enclosure of radius less than $2 \cdot 10^{-25}$. The inverse of the rounded 3×3 matrix is computed exactly, and matrix-perturbation estimates give:

Certified quantity	Rigorous inequality
$G(g_-) - 9/2$	$< -1.0009 \cdot 10^{-10}$
$G(g_+) - 9/2$	$> 2.4365 \cdot 10^{-10}$
Endpoint error in G	$< 2.1 \cdot 10^{-20}$
Secant slope	> 3.43744549
$\|A(g)^{-1}\|_\infty$ on the interval	< 80
$\|G''(g)\|$ on the interval	$< 9 \cdot 10^6$
$G'(g)$ on the interval	> 3.4365

The derivative bounds follow from $T(g) = \int_0^1 e^{-t\widetilde{K}(g)}dt$ and $\|\partial_g T\|_\infty \le 1$, $\|\partial_g^2 T\|_\infty < 4/3$, using the positivity and unit-row-sum property of the heat semigroup after leakage. The unique root g_* therefore lies strictly between the two rational endpoints; no global monotonicity outside that interval is assumed.

Exact local Farkas completion. Let $a(g) = c(g)A(g)^{-1}$. The displayed inequalities establish on the entire interval that all three components $a_i(g) > 1/2$ and that the unheated-w reduced coefficient of the four-guard dual $(a_1, a_2, a_3, 1)$ is $< -1/5$; the other three controlled-column reduced coefficients vanish identically. The dual right-hand side is exactly $G(g) - 9/2$. This is a complete primal–dual characterization of the common first vector letter $(22, 22)$ in the interval. The controls have combined power < 467 wherever $G(g) \le 9/2$, and at the root the closed upper band remains allowed.

All other 80 vector letters. Appendix M prints the complete 81-entry nonnegative rational Farkas family at g_+: 80 previously certified multipliers and one new high-precision symmetrized critical-pair multiplier. For the other 80 pairs, all right-hand sides are > 0.00051730, all held-port reduced coefficients are < -0.00047918, and each multiplier's ℓ_1 norm is < 2. Since the derivative of the first-sample input map has infinity norm at most one, their reduced coefficients change by at most $4(g_+ - g_-) = 4 \cdot 10^{-10}$. Thus the same 80 Farkas witnesses exclude all these letters *throughout* the interval. The new $(22, 22)$ Farkas vector at g_+ has a positive right-side pairing $> 1.88 \cdot 10^{-13}$ and every reduced input coefficient $< -9.4 \cdot 10^{-14}$, using 115 exact positive-series terms and a tail error smaller than 10^{-45}.

All 81 H2 first letters. The right endpoint's 81 exact rational nonnegative twenty-input half-period actions have each global phase power < 421 and minimum first-sample guard margin > 0.09644. They remain valid for every g in the narrow interval, because Duhamel gives a coordinate error at

the first sample of at most $421|g - g_+|$, hence a guard error of at most $842 \cdot 10^{-10}$. This proves a *one-sample* H2 statement; it does not assert an all-horizon controller at these couplings.

Additional explicit primal at the left endpoint. The printed left-endpoint witness certifies one rational H1 held control applied in each module,

$$\begin{aligned} u_r &= 57.316257207698142887177967 26, \quad u_w = 0, \\ u_p &= 76.955659700849080528348534 65, \\ u_{t_1} &= u_{t_2} = 49.538184786128927699334308 44. \end{aligned}$$

The two-cell peak power is less than 467 and the minimum sampled-guard margin is rigorously greater than 10^{-12}. The printed right-endpoint certificate family certifies 81 H1 duals and 81 H2 first-step controls at the opposite endpoint. Only the rational enclosures, algebraic derivative bounds, and Farkas/positive control witnesses certify the phase. The high-precision decimal estimate of g_* is separately labeled exploratory and is *not* asserted with more certified digits than the displayed rational bracket.

L Exact all-horizon two-cross-link certificates

Let K_g be the 12-node leaky generator with the two cross edges r_1w_2 and r_2w_1 each of conductance g. Use the exact rational endpoints

$$g_- = 54757560/10^{10}, \quad g_+ = 54757561/10^{10}, \qquad Q = 421.$$

The main text proves the determinant-root first-sample phase on this interval. The following two independent certificate families complete the phase for all word lengths.

A. H2 symbol-only controller for all 81^n vector words

The complete action table in Appendix M lists 81 rational vectors $u_s \in \mathbb{Q}^{20}_{\geq 0}$ with eight digits of decimal precision: ten first-half and ten second-half heater rates. Each half-phase sum is strictly below 421. Index s by the current pair of three-band terminal words; no previous symbol, future target or measured state is used.

At g_+ let $E = e^{-K_{g_+}}$ and $D = [e^{-K_{g_+}/2}C_{1/2}, C_{1/2}]$ with $C_{1/2} = K_{g_+}^{-1}(I - e^{-K_{g_+}/2})P_{10}$. The exact state after n samples is

$$x_n = \sum_{j=0}^{n-1} E^j D u_{s_{n-j}}.$$

For each of the 18 guard normals h and each lag $j \geq 1$ put $m_{h,j} = \min_{s\in\Sigma^2} hE^jDu_s$. Appendix H prescribes the computation of exact-rational outward matrix-exponential enclosures from 115 non-negative Taylor terms and checks the finite family

$$\min_{s,h}\left[hDu_s + \sum_{j=1}^{k} m_{h,j} - b_{s,h}\right] > 0.0397542, \qquad 0 \leq k \leq 11.$$

The 12-lag cutoff is guaranteed by positivity:

$$\sum_{j=12}^{\infty} |hE^jDu_{s_{n-j}}| \leq 2(421)\sum_{j=12}^{\infty}(3/8)^j = \frac{223736661}{21474836480} < 0.010419.$$

All proxy entries are rational multiples of 10^{-16}, each entry enclosing the exact matrix coefficient with error below 10^{-16}. For the first 12 lags the matrix product row error is below $3 \cdot 10^{-14}$, yielding a guard expression error at most $2(421)(3 \cdot 10^{-14})$ per lag. These errors are *included* in the reported guard margins.

Quantity	Verified bound
81 input vectors, each of 20 coordinates	Nonnegative rationals
Largest first-half power	< 420.999958
Largest second-half power	< 420.999958
Minimum finite-prefix guard margin ($k \le 11$)	> 0.0397542
Bound on all remaining lags	< 0.010419
Uniform all-horizon guard margin at g_+	> 0.02933572
Uniform all-horizon margin throughout $[g_-, g_+]$	> 0.0293355
Margin under $\lvert g(t) - g_+\rvert \le 10^{-5}$	> 0.01249

The last two lines use the global leaky-semigroup Duhamel bound $\|x_{g(\cdot)}(t) - x_{g_+}(t)\|_\infty \le 2Q \sup |g(t) - g_+|$ and a factor two for each signed guard. All horizons are covered without enumerating target sequences.

B. H1, every two-sample vector word impossible

At g_- the exact input–output map for the first joint vector symbol $(22, 22)$ followed by any $t \in \Sigma^2$ is

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} C & 0 \\ EC & C \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, \quad C = K_{g_-}^{-1}(I - e^{-K_{g_-}})P_{10}.$$

The full two-step dual table in Appendix M contains *exact decimal-rational* 36-component nonnegative Farkas vectors y_t for all 81 second letters. Their entries were initially discovered numerically but the final proof checks each integer decimal as an exact fraction. The rational-arithmetic two-step certificate verifies

$$y_t^T b_t > 4.4101 \cdot 10^{-6}, \qquad \max_{i \le 20} (y_t^T A_{g_-})_i < -4.4101 \cdot 10^{-6}.$$

After including the exact matrix-exponential error, the worst coefficient is below $-4.41010665 \cdot 10^{-6}$. The Duhamel variation on $[g_-, g_+]$ is below $7 \cdot 10^{-10}$ per reduced coefficient; hence all 81 certificates retain coefficients below $-4.4094 \cdot 10^{-6}$ uniformly over the bracket. The first letter cannot be anything but $(22, 22)$, by the critical-coupling theorem. Thus every two-sample word is impossible at every finite H1 budget; all longer words are excluded by their first two letters.

C. Diagnostic physical replay

An independent full-network matrix-exponential computation computes the actual 12-dimensional matrix exponential for 37,500 samples across 15 target/coupling protocols, including alternating extreme symbols and conductances switched every half-period. Its lowest observed guard margin is above 0.03972. These finite numerical experiments are diagnostics only; the arbitrary-word result follows from Sections A and B above.

M Full finite certificate tables for the crossed network

All vectors below are printed as exact rationals; no auxiliary data or software is needed to recover their entries. Each listed coordinate is identified by its *one-based* position, and any unlisted coordinate equals exactly zero. In a row with scale 10^{-d}, `index:numerator` means that the entry at that coordinate is the displayed integer divided by 10^d. A parenthesized rational `index:(a/b)` denotes exactly a/b without further scaling. All signed guard rows follow the fixed order of the nine inequalities for each cell: $(r - p,\ p - w,\ b - w,\ p - t_1,\ p - t_2,\ t_1 - b,\ b - t_1,\ t_2 - b,\ b - t_2)$, with lower and upper band right sides as in Section 2; the 18 two-cell rows list cell 1 first, then cell 2. For a

two-sample H1 dual, the 36 entries list the 18 first-sample guards and then the 18 second-sample guards. The twenty heater coordinates in H2 are first the ten cell-major open ports (r, w, p, t_1, t_2) at the early half, then the same ten ports at the late half. All actions are held powers, not energy integrals.

N Eighty-one exact all-horizon H2 controls

This is the memoryless policy of the all-horizon theorem. Each $u_s \in \mathbb{Q}^{20}_{\geq 0}$ is the integer numerator printed in the table times 10^{-8}; use the current vector letter $s = (s_1, s_2)$ to index the row. The two 10-port phase sums are strictly less than 421.

Label	Nonzero coordinates written as `index:numerator`
00/00	4:13102581772 5:6685307180 9:13113964668 10:9198142168 11:5828848705 13:5629309273 15:1355260259 16:5872256544 18:5383928769 20:826980136
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00/02	4:13122192009 5:9465696269 10:19512107511 11:5878721760 13:5361540963 15:772213077 16:7879403195 18:8448038811 19:3941154556 20:1094984174
00/10	4:10799850150 5:13121815266 9:3753129602 10:13611752761 11:5900166445 13:5227695393 14:490374379 16:5776389346 18:5756138548 19:2568087235
00/11	4:12640787094 5:13358693733 6:1556202898 9:11181492558 11:6244514636 13:5169617826 14:151614259 16:5434394232 18:6367023155 19:1692844103 20:4054256614
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01/00	4:13607423669 5:2595895104 9:13152326795 10:12653504610 11:5756659020 13:5869582302 15:2811575191 16:6120367011 18:5153793216
01/01	4:13381383558 9:13610799783 10:3750645172 11:5820921376 13:6255761306 14:164961186 15:3779496180 16:5776518490 18:5755898206 20:2568410661
01/02	1:2053038903 4:3686460353 5:17048334578 9:18605311909 10:706850044 11:5626095406 12:67380861 13:5870490395 14:2335828696 16:7833667542 18:8426949681 20:5054544669
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02/02	1:935756442 4:18898070887 5:1157523413 6:975009318 10:20133635727 11:7620705192 13:8408906847 15:5021201401 16:7620391772 18:8413255765 19:3996482675 20:1019052133
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02/20	1:935756442 4:18898070887 5:1157523413 6:975009318 9:20133635727 11:7620705192 13:8408906847 15:5021201401 16:7620391772 18:8413255765 19:1019052133 20:3996482675
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02/22	1:885795227 5:20030552438 6:8156258448 9:13027389675 11:7629432510 13:8408987816 14:3989053666 15:1033399796 16:4959916138 18:7671615130 19:2828172174 20:5579418557
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Label	Nonzero coordinates written as `index:numerator` (cont.)
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20/22	1:885795227 4:20030552438 6:8156258448 9:13027389675 11:7629432510 13:8408987816 14:1033399796 15:3989053666 16:4959916138 18:7671615130 19:2828172174 20:5579418557
21/00	2:986500466 4:5371374889 9:13354707677 10:12631819408 11:5962082381 13:7297824201 14:3636311186 15:4284631260 16:6241092039 18:5168914709 20:152666327
21/01	4:4253310544 9:13602311072 10:1901058526 11:6150630032 13:7615395070 14:4000354584 15:4307211967 16:5744956376 18:5936554290 20:2957230998
21/02	1:3488870706 3:2910492180 4:9669859006 9:18523792528 10:500507494 11:5701748095 13:7347686107 14:3124165049 15:4680279331 16:7768291285 18:8396919922 20:5080905997
21/10	5:4320204528 9:15912033270 10:1448520912 11:5930185177 13:7294229051 14:4672275226 15:3273836627 16:5776808403 18:5755847514 20:2568489549
21/11	5:4319082463 9:15162350099 11:5930187190 13:7294147948 14:4672210072 15:3274008441 16:5728375656 18:5860935433 19:266138916 20:3468266273

Label	Nonzero coordinates written as `index:numerator` (cont.)
21/12	5:8053361904 9:4323094903 11:6337166744 13:7506228037 14:4753320331 15:2264006941 16:5930701602 18:7294467259 19:3273358947 20:4672401388
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21/21	5:4330458435 6:1488621373 10:6283168812 11:5930977899 13:7294704117 14:4672961615 15:3272357500 16:5761467364 18:7345249600 19:4817633465 20:3004644226
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22/01	4:2161693230 9:13610951047 10:3751127482 11:6104017614 13:7710272729 14:4550051801 15:4901239086 16:5777022978 18:5755966789 20:2568340748
22/02	1:8101072166 5:12956428340 6:939725909 9:18913714788 10:1189054584 11:4966397842 13:7669851558 14:5574174729 15:2837914626 16:7622625176 18:8411192252 20:5017839603
22/10	4:2230990577 9:1441527042 10:14023382575 11:5883906831 13:7389296085 14:4309219786 15:4780381309 16:6084872698 18:6417062284 19:3551029112
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22/20	1:8101072166 4:12956428340 6:939725910 9:20102769373 11:4966397842 13:7669851558 14:2837914626 15:5574174729 16:7622625176 18:8411192252 19:1023463854 20:3994375749
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22/22	1:6436703150 4:10800971889 6:9734090459 9:15128230291 11:5156093381 13:7613915155 14:3133415746 15:5414466331 16:4780869372 18:7726013676 19:2540150254 20:5735071870

O Eighty-one exact two-sample H1 Farkas duals

The first vector letter is $(22, 22)$ and the row label is the second vector letter t. The 36-component nonnegative multiplier uses exact scale 10^{-14}. For each row, $y_t^T b_t > 4.4101 \cdot 10^{-6}$ and every reduced input coefficient is less than $-4.4101 \cdot 10^{-6}$ at g_- after rigorous exponential error; the uniform drift over $[g_-, g_+]$ is at most $7 \cdot 10^{-10}$.

Label	Nonzero coordinates written as `index:numerator`
00/00	1:1889697863377 3:6011462750779 7:3449429134682 9:3449429134682 10:1842196894516 12:4104731362981 16:3256720888236 18:3256720888236 19:1904935676228 21:6698809417144 25:3512504402318 27:3512504402318 28:8346219614973 29:6312131937065 30:21417091204354 34:10517707214053 36:10517707214054
00/01	1:1725650061209 3:3570979857976 7:3034237487489 9:3034237487489 10:1777590700325 12:5655923581863 16:3244957178547 18:3244957178547 19:8833554098180 20:6902094312158 21:22382545085954 25:10958575183271 27:10958575183271 28:1790236613710 30:6288621508133 34:3298632240938 36:3298632240938
00/02	1:1647147192266 3:3211458940453 7:2884378544478 9:2884378544478 10:1702078276016 12:5416441787384 16:3107230268950 18:3107230268950 19:9161809735682 20:7299477387257 21:23032849323193 25:11255532204119 27:11255532204119 28:1712978354374 30:6012329765589 34:3154573601346 36:3154573601346
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21/20	1:1025891399065 3:3271946159180 7:1873928833956 9:1873928833956 10:946416520463 14:2172532791 16:1544921496807 18:1542748964016 19:1021171858465 21:3538300597848 25:1864610896192 27:1864610896192 28:12098850909823 30:39706272041157 31:10855564262279 34:3056549767766 36:13912114030045
21/21	1:438454931468 7:716756453340 9:716756453340 10:720471262563 14:300830366573 16:949319158673 18:648488792100 19:5618101013407 21:18374822559519 22:5053747280618 25:1375663842124 27:6429411122742 28:9004018336698 30:29429643744572 31:8371774722950 34:1739982618183 36:10111757341132
21/22	1:1124750686198 5:770016197956 7:1252966459248 9:482950261292 10:437746859563 12:1399122572591 16:799096195824 18:799096195824 19:13817480723073 21:45137353205017 22:13124662263327 25:2194666278460 27:15319328541787 28:424275846951 30:1409325948749 34:753580882070 36:753580882070
22/00	1:1549160124361 3:4931473734755 7:2828323259389 9:2828323259389 10:1488173172718 12:2483402932560 16:2580903284827 18:2580903284827 19:1556524759803 21:5452818890733 25:2862844378002 27:2862844378002 28:9826551277440 29:8104207018161 30:24349762703766 34:11856891770633 36:11856891770633
22/01	1:1180572848404 3:3762527788761 7:2156057908367 9:2156057908367 10:1104989095619 12:728528329442 16:1849419690071 18:1849419690071 19:1179415805683 21:4104198161276 25:2159672919376 27:2159672919376 28:11428815439290 29:10043892394477 30:27523993561030 34:13306382770195 36:13306382770195
22/02	1:976692569594 3:3115282750652 7:1784018099228 9:1784018099228 10:961334336781 14:66403358728 16:1520499190483 18:1454095831755 19:971240960344 21:3360209918306 25:1771672217805 27:1771672217805 28:12242615867970 30:40160586955394 32:11045376388565 34:14029828812963 36:2984452424399
22/10	1:1180572848404 3:3762527788761 7:2156057908367 9:2156057908367 10:1104989095619 12:728528329442 16:1849419690071 18:1849419690071 19:1179415805683 21:4104198161276 25:2159672919376 27:2159672919376 28:11428815439290 29:10043892394477 30:27523993561030 34:13306382770195 36:13306382770195
22/11	1:686947262347 3:2192649266754 7:1254509293044 9:1254509293044 10:1049189420308 14:444676187495 16:1376669576539 18:931993389044 19:677184301419 21:2311385399171 25:1224331026683 27:1224331026683 28:13089286864365 29:12163231700194 30:30672939528822 34:14723083232044 36:14723083232044
22/12	1:437746859563 3:1399122572591 7:799096195824 9:799096195824 10:1124750686198 14:770016197956 16:1252966459248 18:482950261292 19:424275846951 21:1409325948749 25:753580882070 27:753580882070 28:13817480723073 30:45137353205017 32:13124662263327 34:15319328541787 36:2194666278460
22/20	1:976692569594 3:3115282750652 7:1784018099228 9:1784018099228 10:961334336781 14:66403358728 16:1520499190483 18:1454095831755 19:971240960344 21:3360209918306 25:1771672217805 27:1771672217805 28:12242615867970 30:40160586955394 31:11045376388565 34:2984452424399 36:14029828812963
22/21	1:437746859563 3:1399122572591 7:799096195824 9:799096195824 10:1124750686198 14:770016197956 16:1252966459248 18:482950261292 19:424275846951 21:1409325948749 25:753580882070 27:753580882070 28:13817480723073 30:45137353205017 31:13124662263327 34:2194666278460 36:15319328541787
22/22	1:159371454105 4:102920701986 5:102920701986 10:1104045392072 13:331597814865 14:881939157472 16:550341342607 19:1885573977785 21:6035447075239 23:1903322478895 25:1970718217858 27:67395738963 28:13227427263386 30:43065931876793 32:13017764107359 34:14305523402995 36:1287759295636

P First-sample H1 duals at the right critical endpoint

The 18 components for each first vector letter are exact fractions. Ordinary rows use scale 10^{-12}, except the specially constructed critical $(22, 22)$ row: entries not divisible by 10^{-12} are printed as their unreduced exact rational fractions. The first 80 rows remain strictly excluding throughout the critical bracket; the right critical row certifies the strict obstruction at g_+.

Label	Nonzero coordinates written as `index:numerator`
00/00	1:60678259969 3:214679144971 7:112321297530 9:112321297530 10:60678259969 12:214679144971 16:112321297530 18:112321297530
00/01	1:70906276745 2:16375506907 3:229814438420 7:118451770793 9:118451770793 10:54155492165 12:191453605516 16:100195569331 18:100195569331

Label	Nonzero coordinates written as `index:numerator` (cont.)
00/02	1:81540285500 2:33401024918 3:245550513809 7:124825587710 9:124825587710 10:47373808925 12:167306149042 16:87588521193 18:87588521193
00/10	1:70906276745 2:16375506907 3:229814438420 7:118451770793 9:118451770793 10:54155492165 12:191453605516 16:100195569331 18:100195569331
00/11	1:84514218035 2:38162422347 3:249951302605 7:126608104721 9:126608104721 10:45477227035 12:160553013126 16:84062803705 18:84062803705
00/12	1:92154274819 2:50394490925 3:261256964682 7:131187405368 9:131187405368 10:40604892789 12:143204152233 16:75005206909 18:75005206909
00/20	1:81540285500 2:33401024918 3:245550513809 7:124825587710 9:124825587710 10:47373808925 12:167306149042 16:87588521193 18:87588521193
00/21	1:92154274819 2:50394490925 3:261256964682 7:131187405368 9:131187405368 10:40604892789 12:143204152233 16:75005206909 18:75005206909
00/22	1:98308071803 2:60246991878 3:270363277785 7:134875870953 9:134875870953 10:36680398947 12:129230255064 16:67709631308 18:67709631308
01/00	1:54155492165 3:191453605516 7:100195569331 9:100195569331 10:70906276745 11:16375506907 12:229814438420 16:118451770793 18:118451770793
01/01	1:92141419776 2:50373909434 3:261237941948 7:131179700307 9:131179700307 10:40613090903 12:143233343160 16:75020447082 18:75020447082
01/02	1:100841451557 2:64303044832 3:274112141992 7:136394329241 9:136394329241 10:35064773137 12:123477516023 16:64706206989 18:64706206989
01/10	1:92141419776 2:50373909434 3:261237941948 7:131179700307 9:131179700307 10:40613090903 12:143233343160 16:75020447082 18:75020447082
01/11	1:103233144246 2:68132250534 3:277651339423 7:137827863073 9:137827863073 10:33539506175 12:118046516627 16:61870758424 18:61870758424
01/12	1:109296688073 2:77840252192 3:286624097014 7:141462232680 9:141462232680 10:29672569962 12:104277564203 16:54682181598 18:54682181598
01/20	1:100841451557 2:64303044832 3:274112141992 7:136394329241 9:136394329241 10:35064773137 12:123477516023 16:64706206989 18:64706206989
01/21	1:109296688073 2:77840252192 3:286624097014 7:141462232680 9:141462232680 10:29672569962 12:104277564203 16:54682181598 18:54682181598
01/22	1:114097889872 2:85527188472 3:293728856395 7:144339979159 9:144339979159 10:26610673878 12:93375108305 16:48990162380 18:48990162380
02/00	1:47373808925 3:167306149042 7:87588521193 9:87588521193 10:81540285500 11:33401024918 12:245550513809 16:124825587710 18:124825587710
02/01	1:35064773137 3:123477516023 7:64706206989 9:64706206989 10:100841451557 11:64303044832 12:274112141992 16:136394329241 18:136394329241
02/02	1:27526969899 3:96637752328 7:50693543036 9:50693543036 10:112661093200 12:374829515846 14:83226813498 16:143478791328 18:60251977830
02/10	1:35064773137 3:123477516023 7:64706206989 9:64706206989 10:100841451557 11:64303044832 12:274112141992 16:136394329241 18:136394329241
02/11	1:114494176933 3:380476938685 5:86161661571 7:144577505874 9:58415844303 10:26357947613 12:92475228987 16:48520348017 18:48520348017
02/12	1:119157889455 3:394845052997 5:93628471442 7:147372843968 9:53744372526 10:23383733300 12:81884980499 16:42991327906 18:42991327906
02/20	1:27526969899 3:96637752328 7:50693543036 9:50693543036 10:112661093200 12:374829515846 13:83226813498 16:60251977830 18:143478791328
02/21	1:119157889455 3:394845052997 5:93628471442 7:147372843968 9:53744372526 10:23383733300 12:81884980499 16:42991327906 18:42991327906
02/22	1:122867446130 3:406273573536 5:99567635630 7:149596279705 9:50028644074 10:21018017895 12:73461406624 16:38593498203 18:38593498203
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10/01	1:92141419776 2:50373909434 3:261237941948 7:131179700307 9:131179700307 10:40613090903 12:143233343160 16:75020447082 18:75020447082
10/02	1:100841451557 2:64303044832 3:274112141992 7:136394329241 9:136394329241 10:35064773137 12:123477516023 16:64706206989 18:64706206989
10/10	1:92141419776 2:50373909434 3:261237941948 7:131179700307 9:131179700307 10:40613090903 12:143233343160 16:75020447082 18:75020447082
10/11	1:103233144246 2:68132250534 3:277651339423 7:137827863073 9:137827863073 10:33539506175 12:118046516627 16:61870758424 18:61870758424
10/12	1:109296688073 2:77840252192 3:286624097014 7:141462232680 9:141462232680 10:29672569962 12:104277564203 16:54682181598 18:54682181598
10/20	1:100841451557 2:64303044832 3:274112141992 7:136394329241 9:136394329241 10:35064773137 12:123477516023 16:64706206989 18:64706206989
10/21	1:109296688073 2:77840252192 3:286624097014 7:141462232680 9:141462232680 10:29672569962 12:104277564203 16:54682181598 18:54682181598
10/22	1:114097889872 2:85527188472 3:293728856395 7:144339979159 9:144339979159 10:26610673878 12:93375108305 16:48990162380 18:48990162380
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11/01	1:33539506175 3:118046516627 7:61870758424 9:61870758424 10:103233144246 11:68132250534 12:277651339423 16:137827863073 18:137827863073
11/02	1:26357947613 3:92475228987 7:48520348017 9:48520348017 10:114494176933 12:380476938685 14:86161661571 16:144577505874 18:58415844303
11/10	1:33539506175 3:118046516627 7:61870758424 9:61870758424 10:103233144246 11:68132250534 12:277651339423 16:137827863073 18:137827863073

Label	Nonzero coordinates written as `index:numerator` (cont.)
11/11	1:19280262431 3:67273802021 7:35363036677 9:35363036677 10:125592331113 11:103930296920 12:310738189195 16:151229522483 18:151229522483
11/12	1:129352534136 2:109950548150 3:316302491388 7:153483314650 9:153483314650 10:16882248060 12:58735221612 16:30905163677 18:30905163677
11/20	1:26357947613 3:92475228987 7:48520348017 9:48520348017 10:114494176933 12:380476938685 13:86161661571 16:58415844303 18:144577505874
11/21	1:129352534136 2:109950548150 3:316302491388 7:153483314650 9:153483314650 10:16882248060 12:58735221612 16:30905163677 18:30905163677
11/22	1:132270842402 2:114622888665 3:320620968074 7:155232491562 9:155232491562 10:15021139732 12:52108395976 16:27445391014 18:27445391014
12/00	1:40604892789 3:143204152233 7:75005206909 9:75005206909 10:92154274819 11:50394490925 12:261256964682 16:131187405368 18:131187405368
12/01	1:29672569962 3:104277564203 7:54682181598 9:54682181598 10:109296688073 11:77840252192 12:286624097014 16:141462232680 18:141462232680
12/02	1:23383733300 3:81884980499 7:42991327906 9:42991327906 10:119157889455 12:394845052997 14:93628471442 16:147372843968 18:53744372526
12/10	1:29672569962 3:104277564203 7:54682181598 9:54682181598 10:109296688073 11:77840252192 12:286624097014 16:141462232680 18:141462232680
12/11	1:16882248060 3:58735221612 7:30905163677 9:30905163677 10:129352534136 11:109950548150 12:316302491388 16:153483314650 18:153483314650
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12/21	1:139616474211 3:457874513543 5:126383569848 7:159635319636 9:33251749788 10:10336570698 12:35428108375 16:18736846951 18:18736846951
12/22	1:141352947970 3:463224297291 5:129163741082 7:160676128053 9:31512386971 10:9229160007 12:31484964687 16:16678186969 18:16678186969
20/00	1:47373808925 3:167306149042 7:87588521193 9:87588521193 10:81540285500 11:33401024918 12:245550513809 16:124825587710 18:124825587710
20/01	1:35064773137 3:123477516023 7:64706206989 9:64706206989 10:100841451557 11:64303044832 12:274112141992 16:136394329241 18:136394329241
20/02	1:27526969899 3:96637752328 7:50693543036 9:50693543036 10:112661093200 12:374829515846 14:83226813498 16:143478791328 18:60251977830
20/10	1:35064773137 3:123477516023 7:64706206989 9:64706206989 10:100841451557 11:64303044832 12:274112141992 16:136394329241 18:136394329241
20/11	1:114494176933 3:380476938685 4:86161661571 7:58415844303 9:144577505874 10:26357947613 12:92475228987 16:48520348017 18:48520348017
20/12	1:119157889455 3:394845052997 4:93628471442 7:53744372526 9:147372843968 10:23383733300 12:81884980499 16:42991327906 18:42991327906
20/20	1:27526969899 3:96637752328 7:50693543036 9:50693543036 10:112661093200 12:374829515846 13:83226813498 16:60251977830 18:143478791328
20/21	1:119157889455 3:394845052997 4:93628471442 7:53744372526 9:147372843968 10:23383733300 12:81884980499 16:42991327906 18:42991327906
20/22	1:122867446130 3:406273573536 4:99567635630 7:50028644074 9:149596279705 10:21018017895 12:73461406624 16:38593498203 18:38593498203
21/00	1:40604892789 3:143204152233 7:75005206909 9:75005206909 10:92154274819 11:50394490925 12:261256964682 16:131187405368 18:131187405368
21/01	1:29672569962 3:104277564203 7:54682181598 9:54682181598 10:109296688073 11:77840252192 12:286624097014 16:141462232680 18:141462232680
21/02	1:23383733300 3:81884980499 7:42991327906 9:42991327906 10:119157889455 12:394845052997 14:93628471442 16:147372843968 18:53744372526
21/10	1:29672569962 3:104277564203 7:54682181598 9:54682181598 10:109296688073 11:77840252192 12:286624097014 16:141462232680 18:141462232680
21/11	1:16882248060 3:58735221612 7:30905163677 9:30905163677 10:129352534136 11:109950548150 12:316302491388 16:153483314650 18:153483314650
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21/20	1:23383733300 3:81884980499 7:42991327906 9:42991327906 10:119157889455 12:394845052997 13:93628471442 16:53744372526 18:147372843968
21/21	1:139616474211 3:457874513543 4:126383569848 7:33251749788 9:159635319636 10:10336570698 12:35428108375 16:18736846951 18:18736846951
21/22	1:141352947970 3:463224297291 4:129163741082 7:31512386971 9:160676128053 10:9229160007 12:31484964687 16:16678186969 18:16678186969
22/00	1:36680398947 3:129230255064 7:67709631308 9:67709631308 10:98308071803 11:60246991878 12:270363277785 16:134875870953 18:134875870953
22/01	1:26610673878 3:93375108305 7:48990162380 9:48990162380 10:114097889872 11:85527188472 12:293728856395 16:144339979159 18:144339979159
22/02	1:21018017895 3:73461406624 7:38593498203 9:38593498203 10:122867446130 12:406273573536 14:99567635630 16:149596279705 18:50028644074
22/10	1:26610673878 3:93375108305 7:48990162380 9:48990162380 10:114097889872 11:85527188472 12:293728856395 16:144339979159 18:144339979159
22/11	1:15021139732 3:52108395976 7:27445391014 9:27445391014 10:132270842402 11:114622888665 12:320620968074 16:155232491562 18:155232491562
22/12	1:9229160007 3:31484964687 7:16678186969 9:16678186969 10:141352947970 12:463224297291 14:129163741082 16:160676128053 18:31512386971

Label	Nonzero coordinates written as `index:numerator` (cont.)
22/20	1:21018017895 3:73461406624 7:38593498203 9:38593498203 10:122867446130 12:406273573536 13:99567635630 16:50028644074 18:149596279705
22/21	1:9229160007 3:31484964687 7:16678186969 9:16678186969 10:141352947970 12:463224297291 13:129163741082 16:31512386971 18:160676128053
22/22	1:(31246001677865809/400000000000000000) 3:(253766158128628627/1000000000000000000) 5:(154419493535266239/2000000000000000000) 7:(84059418838353439/1000000000000000000) 9:(13699344141440577/2000000000000000000) 10:(31246001677865809/400000000000000000) 12:(253766158128628627/1000000000000000000) 14:(154419493535266239/2000000000000000000) 16:(84059418838353439/1000000000000000000) 18:(13699344141440577/2000000000000000000)

Q First-sample H2 primal controls at the right critical endpoint

The first 10 coordinates are the early control phase and the last 10 the late phase. Every exact entry is a multiple of 10^{-9}, and each half-phase total is less than 421. These controls certify all 81 first-letter pairs at g_+ and throughout the narrow critical bracket by the stated Duhamel estimate.

Label	Nonzero coordinates written as `index:numerator`
00/00	4:129787204019 5:9483735295 9:9483735295 10:129787204019 11:59650516008 13:64538317461 15:25406815289 16:59650516008 18:64538317461 19:25406815289
00/01	4:129782608580 5:9471614609 10:113816256706 11:59650544930 13:64537108553 15:25408404542 16:59083615046 18:65692304686 19:28723055776 20:10600237838
00/02	4:129804047428 5:9529356485 10:90724436776 11:59658191024 13:64543158402 15:25400737727 16:64791215814 18:75658955486 19:33211064353 20:27850283011
00/10	4:129782608580 5:9471614609 10:113816256706 11:59650544930 13:64537108553 15:25408404542 16:59083615046 18:65692304686 19:34637022826 20:4686270787
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Label	Nonzero coordinates written as `index:numerator` (cont.)
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R First-sample H1 duals at $g = 7/1000$

Exactly 81 nonnegative multipliers for $g = 7/1000$. Each 18-vector is printed in integer units of 10^{-12}; the corresponding right-side pairing is strictly positive and the reduced input coefficients strictly negative using the rational-exponential error enclosure.

Label	Nonzero coordinates written as `index:numerator`
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Label	Nonzero coordinates written as `index:numerator` (cont.)
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Label	Nonzero coordinates written as `index:numerator` (cont.)
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22/22	1:153769087237 3:500850243630 4:151131490102 7:15343160753 9:166474650855 10:2524774690 12:6630999948 13:1637796393 14:1637796393

S First-sample H2 controls at $g = 7/1000$

These 81 independent one-sample witnesses for $g = 7/1000$ are exact decimal rationals at scale 10^{-9}. Each ten-port half-phase control has total power below 421; the guard inequalities are strict.

Label	Nonzero coordinates written as `index:numerator`
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00/02	4:129804047428 5:9529356485 10:90724436776 11:59658191024 13:64543158402 15:25400737727 16:64791215814 18:75658955486 19:33211064353 20:27850283011
00/10	4:129782608580 5:9471614609 10:113816256706 11:59650544930 13:64537108553 15:25408404542 16:59083615046 18:65692304686 19:34637022826 20:4686270787

Label	Nonzero coordinates written as `index:numerator` (cont.)
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00/20	4:129804047428 5:9529356485 10:90724436776 11:59658191024 13:64543158402 15:25400737727 16:64791215814 18:75658955486 19:47010320804 20:14051026560
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01/02	4:113878721305 10:90707720651 11:59091290062 13:65697145627 14:4676636076 15:34640579975 16:64791244736 18:75657746579 19:33210093846 20:27852842771
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Label	Nonzero coordinates written as `index:numerator` (cont.)
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20/22	1:104024441422 4:227919916781 10:32423706555 11:53000966026 13:79228912663 14:9036667820 15:43371677886 16:63510309433 18:78405428469 19:50146790137 20:43299247567
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21/01	4:65253038043 10:113862005180 11:64224343774 13:76811733803 14:34541977761 15:40437437180 16:59091318984 18:65695936719 19:28725642417 20:10593162887
21/02	4:65332218767 10:90770185250 11:64231989867 13:76817783652 14:34529783290 15:40441964837 16:64798919752 18:75662587520 19:33213650995 20:27843208060
21/10	4:65253038043 10:113862005180 11:64224343774 13:76811733803 14:34541977761 15:40437437180 16:59091318984 18:65695936719 19:34639609467 20:4679195837
21/11	4:65236321918 10:88407322572 11:64224372696 13:76810524896 14:34544537521 15:40436466672 16:58524418021 18:66849923944 19:35952985751 20:17282297876
21/12	1:74227836062 4:163196174254 10:66051230640 11:55813425760 13:79362517223 14:21110284662 15:47690294552 16:64256413684 18:76851535821 19:40489778759 20:34425749538
21/20	4:65332218767 10:90770185250 11:64231989867 13:76817783652 14:34529783290 15:40441964837 16:64798919752 18:75662587520 19:47012907445 20:14043951610
21/21	1:74227836062 4:163196174254 10:66051230640 11:55813425760 13:79362517223 14:21110284662 15:47690294552 16:64256413684 18:76851535821 19:48375068160 20:26540460138
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22/02	1:104024441422 4:168564661654 10:91778961682 11:51678129629 13:81924370305 14:24638676688 15:60237743518 16:64833145830 18:75709970827 19:33280724505 20:27697238698
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22/20	1:104024441422 4:168564661654 10:91778961682 11:51678129629 13:81924370305 14:24638676688 15:60237743518 16:64833145830 18:75709970827 19:47079980955 20:13897982248
22/21	1:141911282127 4:214948205506 10:63140512367 11:45795364618 13:81631631212 14:17665553705 15:63060332525 16:62692860390 18:75290790715 19:47541681030 20:26321785804

Label	Nonzero coordinates written as `index:numerator` (cont.)
22/22	1:162357141269 4:235063077854 10:22579780877 11:40436188556 13:79281033441 14:13732456344 15:63375282638 16:58903376283 18:73788811335 19:47625730694 20:42857120710

T First-sample H1 duals at $g = 1/100$

Exactly 81 nonnegative multipliers for $g = 1/100$. Each 18-vector is printed in integer units of 10^{-12}; the corresponding right-side pairing is strictly positive and the reduced input coefficients strictly negative using the rational-exponential error enclosure.

Label	Nonzero coordinates written as `index:numerator`
00/00	1:60733111416 3:214642267055 7:112312310765 9:112312310765 10:60733111416 12:214642267055 16:112312310765 18:112312310765
00/01	1:70981696035 2:16342756338 3:229922410211 7:118512328380 9:118512328380 10:54184627355 12:191286542309 16:100128655497 18:100128655497
00/02	1:81629976894 2:33322882470 3:245798481096 7:124954147553 9:124954147553 10:47380751494 12:167019941016 16:87469835962 18:87469835962
00/10	1:70981696035 2:16342756338 3:229922410211 7:118512328380 9:118512328380 10:54184627355 12:191286542309 16:100128655497 18:100128655497
00/11	1:84605280817 2:38067407579 3:250234515102 7:126754097245 9:126754097245 10:45479637239 12:160239455510 16:83932754631 18:83932754631
00/12	1:92243618025 2:50247770865 3:261622905827 7:131375010962 9:131375010962 10:40599009198 12:142832280960 16:74852196601 18:74852196601
00/20	1:81629976894 2:33322882470 3:245798481096 7:124954147553 9:124954147553 10:47380751494 12:167019941016 16:87469835962 18:87469835962
00/21	1:92243618025 2:50247770865 3:261622905827 7:131375010962 9:131375010962 10:40599009198 12:142832280960 16:74852196601 18:74852196601
00/22	1:98390547987 2:60049883285 3:270787680600 7:135093678022 9:135093678022 10:36671337781 12:128823907165 16:67544643569 18:67544643569
01/00	1:54184627355 3:191286542309 7:100128655497 9:100128655497 10:70981696035 11:16342756338 12:229922410211 16:118512328380 18:118512328380
01/01	1:92312674051 2:50357890052 3:261725865016 7:131416787324 9:131416787324 10:40554884834 12:142674907664 16:74770101868 18:74770101868
01/02	1:101010258092 2:64227365886 3:274693541451 7:136678506504 9:136678506504 10:34997435431 12:122853792226 16:64430296953 18:64430296953
01/10	1:92312674051 2:50357890052 3:261725865016 7:131416787324 9:131416787324 10:40554884834 12:142674907664 16:74770101868 18:74770101868
01/11	1:103398443298 2:68035650675 3:278254209893 7:138123270917 9:138123270917 10:33471469326 12:117411304470 16:61591190252 18:61591190252
01/12	1:109447626615 2:77681892832 3:287273249156 7:141782804845 9:141782804845 10:29606254566 12:103625687650 16:54399839746 18:54399839746
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01/21	1:109447626615 2:77681892832 3:287273249156 7:141782804845 9:141782804845 10:29606254566 12:103625687650 16:54399839746 18:54399839746
01/22	1:114231901802 2:85311067645 3:294406371677 7:144677115771 9:144677115771 10:26549271482 12:92722697771 16:48712229041 18:48712229041
02/00	1:47380751494 3:167019941016 7:87469835962 9:87469835962 10:81629976894 11:33322882470 12:245798481096 16:124954147553 18:124954147553
02/01	1:34997435431 3:122853792226 7:64430296953 9:64430296953 10:101010258092 11:64227365886 12:274693541451 16:136678506504 18:136678506504
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02/10	1:34997435431 3:122853792226 7:64430296953 9:64430296953 10:101010258092 11:64227365886 12:274693541451 16:136678506504 18:136678506504
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02/22	1:123060811832 3:406959809356 5:99389960557 7:150018282355 9:50628321799 10:20907909405 12:72602300649 16:38216302023 18:38216302023
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10/02	1:101010258092 2:64227365886 3:274693541451 7:136678506504 9:136678506504 10:34997435431 12:122853792226 16:64430296953 18:64430296953

Label	Nonzero coordinates written as `index:numerator` (cont.)
10/10	1:92312674051 2:50357890052 3:261725865016 7:131416787324 9:131416787324 10:40554884834 12:142674907664 16:74770101868 18:74770101868
10/11	1:103398443298 2:68035650675 3:278254209893 7:138123270917 9:138123270917 10:33471469326 12:117411304470 16:61591190252 18:61591190252
10/12	1:109447626615 2:77681892832 3:287273249156 7:141782804845 9:141782804845 10:29606254566 12:103625687650 16:54399839746 18:54399839746
10/20	1:101010258092 2:64227365886 3:274693541451 7:136678506504 9:136678506504 10:34997435431 12:122853792226 16:64430296953 18:64430296953
10/21	1:109447626615 2:77681892832 3:287273249156 7:141782804845 9:141782804845 10:29606254566 12:103625687650 16:54399839746 18:54399839746
10/22	1:114231901802 2:85311067645 3:294406371677 7:144677115771 9:144677115771 10:26549271482 12:92722697771 16:48712229041 18:48712229041
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11/01	1:33471469326 3:117411304470 7:61591190252 9:61591190252 10:103398443298 11:68035650675 12:278254209893 16:138123270917 18:138123270917
11/02	1:26225483252 3:91567879540 7:48109811117 9:48109811117 10:114738640620 12:381281026032 14:86119131327 16:144983674160 18:58864542833
11/10	1:33471469326 3:117411304470 7:61591190252 9:61591190252 10:103398443298 11:68035650675 12:278254209893 16:138123270917 18:138123270917
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12/01	1:29606254566 3:103625687650 7:54399839746 9:54399839746 10:109447626615 11:77681892832 12:287273249156 16:141782804845 18:141782804845
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12/21	1:10100338798 3:34056181899 7:18108485723 9:18108485723 10:139975001566 12:459150013359 13:126361927104 16:33888819362 18:160250746466
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20/02	1:112913799085 3:375650318711 4:83209174327 7:60670537669 9:143879711996 10:27391492623 12:95726551023 16:50279207283 18:50279207283
20/10	1:34997435431 3:122853792226 7:64430296953 9:64430296953 10:101010258092 11:64227365886 12:274693541451 16:136678506504 18:136678506504
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21/00	1:40599009198 3:142832280960 7:74852196601 9:74852196601 10:92243618025 11:50247770865 12:261622905827 16:131375010962 18:131375010962
21/01	1:29606254566 3:103625687650 7:54399839746 9:54399839746 10:109447626615 11:77681892832 12:287273249156 16:141782804845 18:141782804845
21/02	1:23261982823 3:80998303488 7:42596127754 9:42596127754 10:119376612498 12:395591891777 14:93515005697 16:147789476953 18:54274471256
21/10	1:29606254566 3:103625687650 7:54399839746 9:54399839746 10:109447626615 11:77681892832 12:287273249156 16:141782804845 18:141782804845
21/11	1:16721907695 3:57672569920 7:30428117567 9:30428117567 10:129612036869 11:109836776237 12:317337407944 16:153981533101 18:153981533101

Label	Nonzero coordinates written as `index:numerator` (cont.)
21/12	1:10100338798 3:34056181899 7:18108485723 9:18108485723 10:139975001566 12:459150013359 14:126361927104 16:160250746466 18:33888819362
21/20	1:23261982823 3:80998303488 7:42596127754 9:42596127754 10:119376612498 12:395591891777 13:93515005697 16:54274471256 18:147789476953
21/21	1:10100338798 3:34056181899 7:18108485723 9:18108485723 10:139975001566 12:459150013359 13:126361927104 16:33888819362 18:160250746466
21/22	1:141658693750 3:464345192077 4:129046802345 7:32222516108 9:161269318453 10:9024518883 12:30219178776 16:16106889804 18:16106889804
22/00	1:36671337781 3:128823907165 7:67544643569 9:67544643569 10:98390547987 11:60049883285 12:270787680600 16:135093678022 18:135093678022
22/01	1:26549271482 3:92722697771 7:48712229041 9:48712229041 10:114231901802 11:85311067645 12:294406371677 16:144677115771 18:144677115771
22/02	1:20907909405 3:72602300649 7:38216302023 9:38216302023 10:123060811832 12:406959809356 14:99389960557 16:150018282355 18:50628321799
22/10	1:26549271482 3:92722697771 7:48712229041 9:48712229041 10:114231901802 11:85311067645 12:294406371677 16:144677115771 18:144677115771
22/11	1:14881447769 3:51108413235 7:27003885476 9:27003885476 10:132492414848 11:114429929070 12:321631911965 16:155724056081 18:155724056081
22/12	1:9024518883 3:30219178776 7:16106889804 9:16106889804 10:141658693750 12:464345192077 14:129046802345 16:161269318453 18:32222516108
22/20	1:20907909405 3:72602300649 7:38216302023 9:38216302023 10:123060811832 12:406959809356 13:99389960557 16:50628321799 18:150018282355
22/21	1:9024518883 3:30219178776 7:16106889804 9:16106889804 10:141658693750 12:464345192077 13:129046802345 16:32222516108 18:161269318453
22/22	1:3447026915 3:8984069343 4:2225388677 5:2225388677 10:152972384744 12:498396753118 14:149931904799 16:165874494263 18:15942589463

U First-sample H2 controls at $g = 1/100$

These 81 independent one-sample witnesses for $g = 1/100$ are exact decimal rationals at scale 10^{-9}. Each ten-port half-phase control has total power below 421; the guard inequalities are strict.

Label	Nonzero coordinates written as `index:numerator`
00/00	4:129837598946 5:9617425470 9:9617425470 10:129837598946 11:59691271192 13:64551313959 15:25389224217 16:59691271192 18:64551313959 19:25389224217
00/01	4:129831048873 5:9600149759 10:114008888449 11:59691313818 13:64549591091 15:25391489358 16:59124769693 18:65705874903 19:28734204639 20:10570704922
00/02	4:129861638513 5:9682537720 10:90904471112 11:59702231672 13:64558222718 15:25380550103 16:64833938291 18:75671375971 19:33221521965 20:27822719282
00/10	4:129831048873 5:9600149759 10:114008888449 11:59691313818 13:64549591091 15:25391489358 16:59124769693 18:65705874903 19:34648171689 20:4656737872
00/11	4:129824498801 5:9582874047 10:88562752481 11:59691356443 13:64547868224 15:25393754499 16:58558268194 18:66860435846 19:35962053963 20:17258540942
00/12	4:129855088440 5:9665262009 10:65458335145 11:59702274297 13:64556499850 15:25382815244 16:64267436792 18:76825936914 19:40449371289 20:34510555302
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00/21	4:129855088440 5:9665262009 10:65458335145 11:59702274297 13:64556499850 15:25382815244 16:64267436792 18:76825936914 19:48334660690 20:26625265901
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Statements and Declarations

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Data and code availability. No source code, executable programs, separate proof supplements or research datasets accompany this publication. All essential model definitions, rational certificate arrays, interval-error bounds and verification steps are incorporated into the present article. These constitute an explicit specification from which independent readers can construct their own verification programs and repeat the finite exact-arithmetic calculations. The resulting programs are not asserted to be uniquely determined by the mathematical text. No measured-building data are analyzed in this article; the publicly accessible empirical archives cited in the Introduction provide contextual motivation, not direct validation of the presented models.

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